PHYSICS
OF LIVING SYSTEMS

2022 Annual Meeting of the International Physics of Living Systems (iPoLS) network

MUSE
MONTPELLIER UNIVERSITÉ D'EXCELLENCE

POLS logo by Sebastian Lehnert

Dear iPOLS participants,

Welcome to this 2022 edition of iPOLS ! We are pleased to host you in Montpellier, for the second time in the history of the network !

We have organized the meeting around thematic sessions, including 1 devoted to education in physics. We will also have 2 poster days (with 1 flash talk session and 2 poster sessions each day), and time for round tables and discussions.

We hope that iPOLS2022 will trigger fruitful discussions, foster collaborations and open new perspectives for all of you, and in particular for students who constitute more than half of the participants. Most of all, we hope that it will be a great moment of science and scientific sharing.

We also want to thank all our sponsors (Centre National de la Recherche Scientifique, Labex EpigenMed and NUMEV, Université de Montpellier, Montpellier Métropole) and all the people who have made this possible.

We wish you a wonderful time in Montpellier !

Here is a map of Montpellier City Center (from google)

You will find the main locations you need to know to find your way. All of them are within walking distance (< 10 min)

1 – Faculté de Medecine (Tuesday) : 2 Rue de l'École de Médecine, 34090 Montpellier
2 – Le Corum (Wednesday to Friday) : Place Charles de Gaulle, 34000 Montpellier
3 – Musée Fabre (Conference dinner, Thursday) : 39 Bd Bonne Nouvelle, 34000 Montpellier
4 – Montpellier St Roch Train Station (note that the Montpellier Sud de France station is located elsewhere)

FINAL PROGRAM				
	TUE MAY 31	**WED JUNE 1**	**THU JUNE 2**	**FRI JUNE 3**
	Faculté de Medecine	**Corum**	**Corum**	**Corum**
9AM-10AM		Synthetic Biology	Motility	Molecular Structure and Function
10AM-11AM		Poster flash talks	Poster flash talks	
11AM-12AM		Posters Day 1 & Coffee	Posters Day 2 & Coffee	Informal discussion, roundtables & Coffee
12AM-1PM	Registration	Sensing & signaling	Intracellular self organisation	Neurosciences
1PM-2PM	Welcome adress	Lunch	Lunch	Lunch
2PM-3PM	Cell & Tissues mechanics and dynamics I	Physics Education POLS-T session	Cell and Tissue Mechanics and Dynamics II	Gene Structure & Expression
3PM-4PM	Coffee break			
4PM-5PM	Collective Phenomena	Posters Day 1 & Coffee	Posters Day 2 & Coffee	Coffee break
5PM-6PM		Roundtables & IPOLS faculty meeting	Organisms behaviour	Cellular regulation, morphology and growth
6PM -	Welcome cocktail	FREE EVENING	Dinner at the Musee Fabre	End of meeting
	FREE EVENING			

WIFI CONNEXION AT CORUM

Network : IPOLS2022

Password : cnrs2022

Tuesday May 31th

12.00 – 13.15 : Registration

13.15 – 13.30 : Welcome Address

Emmanuel Margeat, CNRS Montpellier, France

Herbert Levine, Northeastern University, USA

13.30 – 15.30 : Session I : Cell and Tissues mechanics and dynamics I

Chair : Corey O'Hern, Yale University, USA

Antoine Fruleux, Université Paris-Saclay, France

Abstract n°1 : "Temporal memory as a source of large-scale fluctuations in growing tissues"

Nicola Galvanetto, University of Zurich, Suisse

Abstract n°2 : "Extreme molecular dynamics in a biomolecular condensate"

Emile Gasser, Institut Curie, France

Abstract n°3 : "Circulating tumor cells deformation and recovery in a microvasculature microfluidic in vitro model"

Christina Jayachandran, Ludwigs Maxmillan University of Munich, Germany

Abstract n°4 : "Influence of mechanical forces during angiogenesis"

Jordan Shivers, Rice University, USA

Abstract n°5 : "Strain-induced critical slowing down in disordered networks"

Luca Costa, Center for Structural Biology (CBS), France

Abstract n°6 : "Model membranes and LLPS mechanics by means of correlative Atomic Force and Confocal microscopies"

Thomas Day, Georgia Institute of Technology, USA

Abstract n°7 : "Cellular organization in lab-evolved and extant multicellular species obeys a maximum entropy law"

Ethan Wold, Georgia Institute of Technology, USA

Abstract n°8 : "Structural damping makes insect exoskeleton insensitive to non-sinusoidal deformations"

15.30 – 16.00 : Coffee break

16.00 – 18.00 : Session II : Collective phenomena

Chair : Marco Cosentino Lagomarsino, University of Milano, Italy

Sucheol Shin, University of Texas at Austin, USA

Abstract n°9 : "Mechanical force derived from transcriptional activity can induce local order and frozen motion in interphase chromosome"

Eslam Abdelaleem, Emory University, USA

Abstract n°10 : "Exploring relative advantages of dual vs single dimensionality reduction"

Linnea Bavik, Emory University, USA

Abstract n°11 : "Robust Altruism by Similarity in Multifaceted Phenotypes"

Zhijie Feng, Boston University, USA

Abstract n°12 : "Modeling multi-trophic ecosystems with consumer resource model"

María-José Franco-Oñate, Laboratoire Charles Coulomb, Univ. Montpellier, France

Abstract n°13 : "Kinetics of stator recruitment and release in the bacterial flagellar motor"

Loïc Marrec, Universität Bern, Suisse

Abstract n°14 : "Evolutionary rescue in a fluctuating environment"

Richard Servajean, École Polytechnique Fédérale de Lausanne, Suisse

Abstract n°15 : "Impact of finite size effects on the exploration of fitness landscapes by asexual microbial populations"

Jiangguo Zhang, Rice University, USA

Abstract n°16 : "Quantification of Myxococcus xanthus Aggregation and Rippling Behaviors with Deep-Learning Approaches

18.00 – 20.00 : Welcome cocktail

Wednesday June 1st

08.30 – 09.00 : Doors open

09.00 – 10.00 : Session III : Synthetic Biology

Chair : Zan Luthey Schulten, UIUC, USA

Troy Brier, University of Illinois at Urbana-Champaign, USA

Abstract n°17 : "Deciphering the genome architecture in a minimized bacterial genome"

Andrew Hecht, Rice University, USA

Abstract n°18 : "Understanding target selection in high-fidelity Cas9 variants"

Selma Piranej, Emory University, USA

Abstract n°19 : "DNA motors as sensors: toward a new platform for SARS-CoV-2 diagnostics"

Zane Thornburg, University of Illinois at Urbana-Champaign, USA

Abstract n°20 : "Fundamental Cell Behavior Emerges from Whole Cell Simulations of a Minimal Cell by Integrating Experiment and Theory"

10.00 – 10.30 : Posters flash talks Day I

10.30 – 12.00 : Posters Day I & Coffee

12.00 – 13.00 : Session IV : Sensing & Signalling

Chair : Arpita Upadhyaya, University of Maryland, USA

Vernita Gordon, University of Texas at Austin, USA - ***ONLINE***

Abstract n°21 : "Bacteria use a two-stage mechanosensing process to initiate biofilms"

Christoph Haselwandter, University of Southern California, USA

Abstract n°22 : "Force-sensing through ion channels"

Ben Machta, Yale University, USA

Abstract n°23 : "Signaling through formation of a pre-wet surface density"

Viraaj Jayaram, Yale University, USA

Abstract n°24 : "Extracting and exploiting multiple temporal features of odor signals to navigate complex plumes"

13.00 – 14.00 : Lunch

14.00 – 16.00 : Session V : Physics Education : POLS-T session

Eric Mazur, Harvard University, USA **- *ONLINE***

Abstract n°25 : "The PoLS-Teacher Network: High-school Teachers and the Future of Our Discipline"

Isaura J. Gallegos, Harvard Universty, USA

Abstract n°26 : "Research-Practice Partnerships to Improve Physics Education"

Maqsuda Afroz, Cesar E Chavez High School, HoustonISD, USA

Abstract n°27 : "The Road Less Traveled- Social Emotional Learning in Physics"

Maya Lerner, Great Neck North High School, Great Neck, NY, USA

Abstract n°28 : "Training the mind to think: The importance of mindset in physics"

Jolene Johnson, Osseo Public Schools, North Hennepin Community College, University of Wisconsin River Falls, USA

Abstract n°29 : "Addressing the lack of diversity in physics-Equity issues and solutions in high school physics education"

Fabien Kunis, Sofia University St. Kliment Ohridski and Boyn Penev High School, Sofia, Bulgaria, Bulgaria

Abstract n°30 : "Think globally, act locally"

Mollie Thurman, BioBus, USA

Abstract n°31 : "Creating Community Scientists"

16.00 – 17.00 : Posters Day I & Coffee

17.00 – 18.30 : Roundtables & IPOLS Faculty meeting

Thursday June 2nd

08.30 – 09.00 : Doors open

09.00 – 10.00 : Session VI : Motility

Chair : Francesco Pedaci, CNRS Montpellier, France

Corey Herr, University of Maryland, USA

Abstract n°32 : "Modelling cell motility on asymmetric nanotopography"

Jean de Seze, Institut Curie, France

Abstract n°33 : "Controlling antagonist responses with a single activator"

Anais Biquet-Bisquert, Center for Structural Biology (CBS), France

Abstract n°34 : "Electrical characterization of single bacterial cells"

Johannes Heyn, Ludwig-Maximlians-University Munich, Germany

Abstract n°35 : "On the adhesion-velocity relation and multistability of the motile state of MDA MB 231 cells on fibronectin lanes"

10.00 – 10.30 : Posters flash talks Day II

10.30 – 12.00 : Posters Day II & Coffee

12.00 – 13.00 : Session VII : Intracellular organization

Chair : Herbert Levine, Northeastern University, USA

Jean-Charles Walter, CNRS & UM, France

Abstract n°36 : "Genomic physics: biophysical modeling confronted to genomic data"

Diana Valverde Mendez, Princeton University, USA

Abstract n°37 : "3D diffusion in live Escherichia coli cells."

Aradhya Rajanala, Georgia Institute of Technology, USA

Abstract n°38 : "Modeling plant root circumnutation using cellular simulation"

Laura Troyer, University of Illinois at Urbana-Champaign, USA

Abstract n°39 : "Membrane-binding domain affects the diffusion of the RNA degradosome in Escherichia coli"

13.00 – 14.00 : Lunch

14.00 – 16.00 : Session VIII : Cell and Tissues Mechanics and dynamics II

Chair : Christine Doucet, CNRS Montpellier, France

Daniel Cohen, Princeton University, USA - ***CANCELED***

Abstract n°40 : "Programming collective cell migration with bioelectric cues"

Sadhana Gollapudi, University of Texas at Austin, Austin, USA

Abstract n°41 : "Impact of glycosylation on uptake of receptors by endocytosis"

Majid LAYACHI, Université de Montpellier, France

Abstract n°42 : "Microfluidic flow of vesicle prototissues as a model for cellular tissues"

Le Andy Vinh, University of Ottawa, Canada

Abstract n°43 : "Mixing in microcirculation"

Maryam Kohram, Princeton Unviersity, USA

Abstract n°44 : "Bacterial Growth Mechanisms and Their Role in Cell Size Homeostasis"

Garegin Papoian, University of Maryland, College Park, USA

Abstract n°45 : "Towards Simulating Eukaryotic Cells from the Fundamental Physico-Chemical Principles"

Andrew Schulz, Georgia Institute of Technology, USA

Abstract n°46 : "The Muscular Multitool: Biomechanics of the Elephant Trunk from Macro to Micro"

Sahand Rahi, EPFL, Suisse

Abstract n°73 : "Optimal checkpoint strategies balancing risk and speed"

Mikael Toye, Georgia Institute of Technology, USA

Abstract n°47 : "Quantification of Chaos in Rapidly Paced Cardiac Tissue"

16.00 – 17.00 : Posters Day II & Coffee

17.00 – 18.30 : Session IX : Organisms behaviour

Chair : Daniel Goldman, Georgia Tech, USA

Éric Herbert, Université Paris Cité, France

Abstract n°48 : "Thermodynamics of Animal Locomotion"

Sena Agezo, Emory University, USA

Abstract n°49 : "Measuring animal behavior in social contexts using deep learning approaches"

Grace McKenzie-Smith, Princeton University, USA

Abstract n°50 : "Socially isolated bumblebees have perturbed individual and social behaviors compared to socially reared bumblebees"

Nicola Rigolli, University of Genova - Université Côte d'Azur, Italy

Abstract n°51 : "Alternation emerges as a multi-modal strategy for turbulent odor navigation"

Ahmed Roman, Emory University, USA

Abstract n°52 : "Modeling Learning behavior in C.elegans"

Usama Sikandar, Georgia Institute of Technology, USA

Abstract n°53 : "Body oscillations couple with wing flapping to reduce the aerodynamic power in wild silkmoths"

18.30 – 23.59 : *A night at the Museum.*

Visit and dinner at the Musée Fabre.

Friday June 3rd

08.30 – 09.00 : Doors open

09.00 – 10.30 : Session X : Molecular Structure and function

Chair : Don Lamb, LMU Munchen, Germany

Sergei Sukharev, University of Maryland, College Park, USA

Abstract n°54 : "The biophysics of the bacterial hypoosmotic response"

Grace Meng, Yale University, USA

Abstract n°55 : "Quality assessment of computational models of protein-protein interfaces"

Matteo Paloni, Center for Structural Biology (CBS), France

Abstract n°56 : "Unraveling interactions in biomolecular condensates by atomistic molecular dynamics simulations"

Kevin Ng Chau, Northeastern University, USA

Abstract n°57 : "Contact map dependence of a T cell receptor binding repertoire"

Zachary Smith, University of Maryland at College Park, USA

Abstract n°58 : "Understanding Drug Resistance through Molecular Simulation"

Esteban Dodero Rojas, Rice University, USA

Abstract n°59 : "Sterically confined rearrangements of SARS-CoV-2 Spike protein control cell invasion

10.30 – 11.45 : Roundtables, informal discussions and coffee

11.45 – 13.00 : Session XI : Neurosciences

Chair : Ilya Nemenman, Emory University, USA

Ilya Nemenman, Emory University, USA

Abstract n°60 : "Emergent dynamics in high throughput data"

Amitava Banerjee, University of Maryland, College Park, USA

Abstract n°61 : "Using Machine Learning for Inference of C. elegans Neural Network Structure from Calcium Fluorescence Time Series"

Helena Casademunt, Harvard University, USA

Abstract n°62 : "Neural basis for odor-evoked behavior in C. elegans"

Philipp Fleig, Max Planck Institute for Medical Research, Germany

Abstract n°63 : "Statistical properties of large data sets with linear latent features"

Tyler Salners, University of Illinois at Urbana Champaign, USA

Abstract n°64 : "Recurrent activity in neuronal avalanches"

13.00 – 14.00 : Lunch

14.00 – 16.00 : Session XII : Gene structure & expression

Chair : Erwin Frey, LMU Munchen, Germany

Marcelo Nollmann, Center for Structural Biology (CBS), France

Abstract n°65 : "Monitoring DNA organization and transcription in single cells using multiplexing microscopy"

Marta Biondo, University of Turin, Italy

Abstract n°66 : "Identification of the source of extrinsic noise from the stochastic dynamics of gene expression"

Nicola Dietler, EPFL, Suisse

Abstract n°67 : "Disentangling phylogeny and optimization contributions in protein sequences"

Derek Hart, Georgia Institute of Technology, USA

Abstract n°68 : "Weak tension accelerates hybridization and dehybridization of short oligonucleotides"

Sangjin Kim, University of Illinois Urbana-Champaign, USA

Abstract n°69 : "Traffic on the genome: Group dynamics of RNA polymerases during gene transcription via DNA supercoiling"

Ovidiu Radulescu, University of Montpellier, France

Abstract n°70 : "Unravelling individual transcription events and transcription bursting mechanisms by signal deconvolution in RNA tagging systems"

Maria Ruiz Ortega, ENS Paris, France

Abstract n°71 : "Statistical Analysis of Sharing in Antibody Repertoires"

Michele Di Pierro, Northeastern University, USA

Abstract n°72 : "Landscapes of Genomic Architecture Across Evolution"

16.00 – 16.30 : Coffee Break

16.30 – 18.00 : Session XII : Cellular regulation, morphology and growth

Chair : Catherine Villard, CNRS Paris

Baptiste Alric, LAAS-CNRS, France

Abstract n°74 : "Impact of mechanical constrains on cells rheology and physiology"

Mattia Corigliano, IFOM - FIRC institute for molecular oncology, Italy

Abstract n°75 : "Coordination of chromosome replication/segregation and cell-cycle progression in bacteria: the role of rapid chromosome movements?"

Evelyne Crapez, ICM Montpellier, France

Abstract n°76 : "Characterization of colon tissues stiffness during carcinogenesis by atomic force microscopy"

John Treado, Yale University, USA

Abstract n°77 : "Localized growth drives pore space patterning in mesophyll development"

Rossana Droghetti, IFOM - FIRC institute for molecular oncology, Italy

Abstract n°78 : "A mechanistic theory of bacterial resource allocation out of steady state."

POSTER DAY 1

WEDNESDAY 10h30 – 12h00 and 16h00 – 17h00

POSTERS WITH FLASH TALKS (Wednesday 10h-10h30)

Board n°	Name	Institution	Title	Abstract n°
1	Andrew Ton	Yale University, USA	Hole closure in sheets of active deformable particles	79
2	Rachel Topno	Universite de Montpellier, France	Extrinsic and intrinsic transcriptional noise in the control of HIV-1 latency	80
3	Vishavdeep Vashisht	University of Maryland- College Park, USA	Competition between actin treadmilling and myosin contractility regulates the formation of actomyosin rings	81
4	Lixinhao Yang	Georgia Institute of Technology, USA	Developing a coarse-grained model for secretion of the passenger domain of the autotransporter EspP	82
5	Nathan Zimmerberg	University of Maryland, USA	Towards Cell Scale Simulations of the Mechanics and Chemistry of the Cytoskeleton and Membrane.	83
6	Emma Dawson	Emory University, USA	Searching for the Mechanism of Pattern Formation in Proteus mirabilis	84
7	Davide Erba	CBS - Montpellier, France	Structural and dynamics properties of protein/RNA condensates by multiscale molecular simulations and small-angle scattering experiments	85
8	Arianna Giannetti	Centre de Biologie Structurale (CBS), France	The onset of Thrombosis in Deep Veins (DVT): The coupling effect of stretch and shear stresses on endothelial cells	86
9	Lauren Harkins	Yale University, USA	Evaluating Nanoparticle Delivery during Normothermic Machine Perfusion of Human Liver	87
10	Hana Jaafari	Rice University, USA	Characterizing Evolutionary and Physical Energy Landscapes of Evolving, Devolving, and Random Sequences	88
11	Silvia Lazzardi	Università di Torino, Italy	Emergent statistical laws in single-cell transcriptomic data	89
12	Jacopo Marchi	Georgia Institute of Technology, USA	Multi-strain phage-induced clearance of Pseudomonas aeruginosa	90
13	Matheus Mello	Rice University, USA	Investigating human chromosome organization by whole-genome simulations	91
14	Elissa Moller	University of Maryland, USA	MscS is an essential component of the bacterial osmolyte release system	92
15	Jules Berlin Nde Kengne	University of Washington / Rice-CTBP, USA	Coarse-grained Molecular Dynamics Simulations of Ca2+-Calmodulin	93
16	Spandan Pathak	University of Maryland, College Park., USA	Central Nervous System and serotonin provide distinct guidance cues for hindgut motility	94

17	Leo Bonhomme	CBS Montpellier, France	Double labeling strategy using non-canonical amino acids for accurate single molecule FRET measurements on a G protein coupled receptor	95
18	Pablo Bravo	Georgia Institute of Technology, USA	Universal biofilm height dynamics	96
19	Ludovico Calabrese	FIRC Institute of Molecular Oncology (IFOM), Italy	Transcription-limited growth	97
20	Francesco Camaglia	CNRS, LPENS, France	Probing T cell thymic maturation through repertoire sequencing	98
21	Carole Chevalier	Université de Montpellier, France	Physical modeling of ribosomes along messenger RNA with a ballistic model: application to the estimation of the kinetic parameters from ribosome sequencing experiments	99
32	Matteo Boccalini	CBS / université de Montpellier, France	CONFORMATIONAL ENSEMBLE OF RNA IN BIOMOLECULAR CONDENSATES USING ATOMISTIC MOLECULAR DYNAMICS SIMULATIONS	126

POSTERS WITH REGULAR TALKS DURING THE SESSIONS

Board n°	Name	Institution	Title	Abstract n°
22	Nicola Galvanetto	University of Zurich, Suisse	Extreme molecular dynamics in a biomolecular condensate	2
23	Emile Gasser	Institut Curie, France	Circulating tumor cells deformation and recovery in a microvasculature microfluidic in vitro model	3
24	Christina Jayachandran	Ludwigs Maxmillan University of Munich, Germany	Influence of mechanical forces during angiogenesis	4
25	Zhijie Feng	Boston University, USA	Modeling multi-trophic ecosystems with consumer resource model	12
26	Anais Biquet-Bisquert	Center for Structural Biology (CBS), France	Electrical characterization of single bacterial cells	34
27	Mikael Toye	Georgia Institute of Technology, USA	Quantification of Chaos in Rapidly Paced Cardiac Tissue	47
28	Nicola Rigolli	University of Genova - Université Côte d'Azur, Italy	Alternation emerges as a multi-modal strategy for turbulent odor navigation	51
29	Helena Casademunt	Harvard University, USA	Neural basis for odor-evoked behavior in C. elegans	62
30	Marta Biondo	University of Turin, Italy	Identification of the source of extrinsic noise from the stochastic dynamics of gene expression	66
31	Mattia Corigliano	IFOM - FIRC institute for molecular oncology, Italy	Coordination of chromosome replication/segregation and cell-cycle progression in bacteria: the role of rapid chromosome movements?	75

POSTER DAY 2

THURSDAY 10h30 – 12h00 and 16h00 – 17h00

POSTERS WITH FLASH TALKS (Thursday 10h-10h30)

Board n°	Name	Institution	Title	Abstract n°
1	Lucie Albert	LAAS-CNRS/LIED-CNRS, France	Study of the mechanical impacts on the phenotypic plasticity of Candida albicans in vitro using microfluidics.	100
2	Aleksander Atemin	Institute of Molecular Biology (IMB), Bulgarian Academy of Sciences (BAS), Bulgaria	Dynamic Changes in the Replication Complex During Normal and Perturbed Replication	101
3	Janani Baskaran	Yale University, USA	Developing a 3D spheroid model to study how tumor-associated macrophages evolve in the early melanoma tumor microenvironment	102
4	Ecenaz Bilgen	Ludwig-Maximilians-Universität München, Germany	Investigation of the conformational dynamics of chaperone proteins using single molecule FRET	103
5	Thibault Chassereau	Université Paris Cité, France	Hyphal network and Brownian motion	104
6	Alexander Cook	Georgia Institute of Technology, USA	The role of DNA strand displacement kinetics in CRISPR-Cas r-loop formation	105
7	Christine Doucet	CBS, France	Structure and mechanics of the human Nuclear Pore Complex basket	106
8	Frank Fazekas	University of Maryland, USA	Characterizing Nuclear Deformation in Activated T-Cells	107
9	Philippe Fuchs	University of Strasbourg, France	Interdependence between mRNA translation and growth : the cost of unnecessary protein synthesis	108
10	Madeline Galbraith	Rice University, USA	Identifying phenotypes associated with coupling the epithelial to mesenchymal transition and metabolic reprogramming	109
11	Benjamin Gilbert	University of Illinois at Urbana-Champaign, USA	Dynamics of Chromosome Organization During the Cell Cycle of a Minimal Bacterial Cell	110
12	Daniel Goldman	Georgia Tech, USA	TBD	111
13	DAVIN JEONG	University of Texas at Austin, USA	Local Epigenetic Landscape Controls the Integrity of Topologically Associating Domains upon Cohesin loss	112

14	Davneet Kaur	University of Illinois at Urbana-Champaign, USA	Friend or Foe: The Potential of Transposable Element Tuning	113
15	Johannes Keisers	Centre de Biologie Structuarle de Montpellier, France	Traffic slowdown by antibiotics	114
16	Svilen Kolev	Northeastern University, USA	High Throughput Platform for Imaging Worms' Microbiome Composition Dynamics	115
17	Zan Luthey-Schulten	University of Illinois at Urbana-Champaign, USA	Fundamental behaviors emerge from simulations of a living minimal cell	116
18	Asheesh Momi	Yale, USA	Tuning to criticality in the cochlea	117
19	Antonio Bento Oliveira Junior	Rice University, USA	Chromosome Modeling on Downsampled Hi-C Maps	118
20	Core Francisco Park	Harvard University, USA	Brain Imaging of a Freely Moving C. elegans performing thermotaxis	119
21	Anjana Rajkumar Kammath	Yale University, USA	Exploring the hierarchical structural organization of squid iridocyte protein	120
22	Matthieu Rousset	IBMM, University of Montpellier, France, France	The SuperBeeLive macroscope, a monitoring tool to investigate the bee-environment relationship.	121
23	Jacob Sumner	Yale University, USA	Radius of gyration scaling of proteins compared to polymer model simulations	122
24	Arpita Upadhyaya	University of Maryland, College Park, USA	Transcription factor dynamics in gene expression: the long and short of it	123
25	Filippo Valle	University of Turin, Italy	Multi-omics topic modeling for breast cancer classification	124

POSTERS WITH REGULAR TALKS DURING THE SESSIONS

Board n°	Name	Institution	Title	Abstract n°
26	Lam Vo	Yale University, USA	Emergent property of collective migration: spatial organization of phenotypes	125
27	Eslam Abdelaleem	Emory University, USA	Exploring relative advantages of dual vs single dimensionality reduction	10
28	María-José Franco-Oñate	Laboratoire Charles Coulomb, Univ. Montpellier, France	Kinetics of stator recruitment and release in the bacterial flagellar motor	13
29	Jiangguo Zhang	Rice University, USA	Quantification of Myxococcus xanthus Aggregation and Rippling Behaviors with Deep-Learning Approaches	16
30	Laura Troyer	University of Illinois at Urbana-Champaign, USA	Membrane-binding domain affects the diffusion of the RNA degradosome in Escherichia coli	39

31	Éric Herbert	Université Paris Cité, France	Thermodynamics of Animal Locomotion	48
32	Derek Hart	Georgia Institute of Technology, USA	Weak tension accelerates hybridization and dehybridization of short oligonucleotides	68
33	Sangjin Kim	University of Illinois Urbana-Champaign, USA	Traffic on the genome: Group dynamics of RNA polymerases during gene transcription via DNA supercoiling	69

LIST OF ABSTRACTS

Abstract N° 1

Antoine Fruleux Université Paris-Saclay, France

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Temporal memory as a source of large-scale fluctuations in growing tissues

Antoine Fruleux (1,2,3), Lilan Hong (4), Adrienne H. K. Roeder (5), Chun-Biu Li (6), and Arezki Boudaoud (1,2)

(1) : RDP, Université de Lyon, ENS de Lyon, UCB Lyon 1, INRAE, CNRS, 69364 Lyon Cedex 07, France ; (2) : LadHyX, CNRS, Ecole polytechnique, Institut Polytechnique de Paris, 91128 Palaiseau Cedex, France ; (3) : LPTMS, CNRS, Université Paris-Saclay, 91405, Orsay, France ; (4) : Institute of Nuclear Agricultural Sciences, Key Laboratory of Nuclear Agricultural Sciences of Ministry of Agriculture and Zhejiang Province, College of Agriculture and Biotechnology, Zhejiang University, Hangzhou, China ; (5) Weill Institute for Cell and Molecular Biology and Section of Plant Biology, School of Integrative Plant Science; Cornell University, Ithaca, New York 14853, USA ; (6) : Department of Mathematics, Stockholm University, 106 91 Stockholm, Sweden ;

Many recent studies investigate the impact of fluctuations on the development of tissues. Such studies aim, for example, to understand their role in the self-organization of tissues or to describe how the cellular responses they induce impact the developments of tissues and organs. To better understand the role played by fluctuations during morphogenesis, a first step is to have a better characterization of their properties. Using Cellular Fourier Transforms [1], we dissected the spacial variation of growth in plant tissues and we showed that space correlations may result from the combined effects of growth, and time persistence [2]. Our work shed a new light on the effects of memory during organogenesis.

I will present the harmonic decomposition of growth in plant tissue, and how, with the help of statistical inference, we can quantify spatial heterogeneity in tissues. I will compare divers genotypes and explain what can be learned on the impact of the cells mechanical response on fluctuations. Then, I will explain how we quantified the time persistence of fluctuations and how it possibly sets space heterogeneity. Finally, I will express the questions raised by our findings.

[1] Antoine Fruleux and Arezki Boudaoud. Cellular fourier analysis for geometrically disordered materials. Physical Review Research, 3(2):023036, 2021.

[2] Antoine Fruleux, Lilan Hong, Adrienne H. K. Roeder, Chun-Biu Li, and Arezki Boudaoud.

In prep.

Abstract N° 2

Nicola Galvanetto University of Zurich, Suisse

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Extreme molecular dynamics in a biomolecular condensate

Nicola Galvanetto (1,2), Milos Ivanovic (1), Aritra Chowdhury (1), Daniel Nettels (1), Robert Best (3), Ben Schuler (1,2)

(1) : Department of Biochemistry, University of Zurich, Winterthurerstrasse 190, 8057 Zurich, Switzerland; (2) : Department of Physics, University of Zurich, Winterthurerstrasse 190, 8057 Zurich, Switzerland; (3) : Laboratory of Chemical Physics, National Institute of Diabetes and Digestive and Kidney Diseases, National Institutes of Health, Bethesda, 20892-0520, Maryland, USA.

A wide range of biomolecules in solution can phase-separate and form membraneless organelles in the cell. Such assemblies often have liquid-like properties, and the dynamics of the constituent molecules and their

exchange with the environment are crucial for biological function. The materials properties of such assembles are commonly assessed based on their rheological properties or the translational diffusion of constituent molecules, covering timescales of milliseconds to seconds. However, information on the dynamics at the molecular level is lacking. Using coacervates of two disordered highly oppositely charged proteins as an example of phase-separated assemblies, we show with single-molecule fluorescence spectroscopy that the underlying chain dynamics occur on a sub-microsecond timescale, remarkably close to their chain dynamics in dilute solution. The results from single-molecule spectroscopy are in good agreement with large-scale all-atom molecular dynamics simulations and illustrate the subtle balance of forces required to enable liquid-liquid phase separation on the one hand but maintain exceedingly dynamic molecular behavior on the other.

Abstract N° 3

Emile Gasser Institut Curie, France

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Circulating tumor cells deformation and recovery in a microvasculature microfluidic in vitro model

Emile Gasser (1), Kyohei Terao (2), Jean-Louis Viovy (1), Jean-Yves Pierga (3) and Catherine Villard (1)

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BACKGROUND

Most Circulating Tumor Cells (CTCs) die in the blood circulation, which imposes shear stress as well as large and repetitive deformations. Unfortunately, some survive, and eventually seed metastases. It was recently proposed that the blood stream was not a passive step for CTCs dissemination but instead induces both a selection of the most aggressive phenotypes as well as DNA damages and changes in gene expression. We developed a microfluidic metastasis-on-chip device to elucidate how CTCs react to theses harsh mechanical conditions. This in vitro model allow to observe the short-term dynamics of single cell deformation and recovery, extending the knowledge of cellular biomechanics to the very fast deformations induced by forced migration under flow in addition to the knowledge already available on spontaneous migration.

METHODS

We implemented flow-induced forced cell deformation into constrictions mimicking the microvasculature. The exiting cell is then trapped in a nest structure allowing for its observation during recovery in a pressure-free relaxation chamber. This set-up is well-suited to decipher the fast dynamics of both cell nucleus and cytoplasm recovery, as well as DNA damages and nuclear envelope ruptures.

Breast cancer cell lines (MDA-MB-231 and SK-BR-3) of different metastatic potential were used as model for CTCs. Cytoplasmic membrane and nucleus were fluorescently labeled, and their shape evolution through the device was followed.

RESULTS

Both cell types display similar recovery dynamics (≈10s) at the cytoplasmic membrane level. However, nuclei from SK-BR-3 cells, which are stiffer than MDA-MB-231 cells and stem from a less aggressive cancer, exhibit a faster dynamic of recovery (≤20ms) compared to MDA-MB-231 nuclei, and sometime display nuclear membrane rupture. Thus, the nucleus behavior seems discriminant in the cell ability to sustain capillary-like fast deformations.

Interestingly, recovery of the cytoplasmic membrane was better fitted with 2 characteristic times, whose links with mechanical properties of the cell remain unknown, but express complex dynamic of the whole cell rheology.

CONCLUSION

Nucleus behavior during fast deformations may be linked to CTCs metastatic potential. Understanding how metastasis seeding CTCs survive the blood circulation may provide essential data to identify new bi-omarkers of cancer cells adaptation to circulation, and new targets for drugs aimed at this critical step of cancer dissemination.

Abstract N° 4

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Influence of mechanical forces during angiogenesis

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Angiogenesis, a process where new blood vessels form from existing ones occurs during developmental and pathological conditions. Understanding this fundamental process, has important implications in the progress of tumour invasion and in treating medical conditions such as graft rejection, psoriasis and ocular disorders. The quiescent endothelial cells in the presence of VEGF (vascular endothelial growth factor), when activated, invade into their surrounding matrix and forms tubes or sprouts. A commonly used model to understand such migratory behaviour is the classical angiogenesis model where spheroids are embedded in a 3D collagen gel. In a biological driven process, the self-aggregated endothelial cells in the spheroid, migrate into their surrounding matrix by degrading it to form a new three-dimensional (3D) capillary network. The growth of such tubular networks is reported to be influenced by the ECM's mechanical properties. For example, the stress experienced by the ECM is hypothesized to impact the VEGF receptor formation, thereby triggering the migration of endothelial cells. We aim to investigate if such migration is a biochemically or mechanically driven process. While mechanical forces during tube elongation have previously been measured by traction force microscopy (TFM), we in this work, aim to investigate the role of mechanical forces in tube initiation, formation and maintenance. We approach it qualitatively and quantitatively. Qualitatively we use time-lapse (phase contrast) and confocal microscopy to understand the nature of invading tubes into the collagen matrix. Quantitatively, we first aim to determine the ECM displacement field through TFM. Later, we aim to investigate local forces surrounding the 3D tubes via our DNA based molecular force sensor through fluorescence lifetime imaging microscopy (FLIM).

Abstract N° 5

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Strain-induced critical slowing down in disordered networks

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Common biological materials contain embedded fibrous networks that stiffen significantly in response to applied strain, a phenomenon that provides living tissues with enhanced mechanical resilience and enables long-range force transmission by cells. Prior work has shown that strain-induced stiffening in mechanically under-constrained elastic networks corresponds to the traversal of a "critical point" in strain, at which the energy-minimizing network configuration becomes marginally stable. Near this transition, various quantities measured via quasistatic deformation exhibit power law scaling with respect to the reduced strain. However, the dynamical response of systems near this transition remain poorly understood. Here, we model the small-amplitude oscillatory shear and stress relaxation of under-constrained, fluid-immersed elastic networks subjected

to applied extensional pre-strain (e.g. swelling and simple shear). We show that the rheology of these networks is controlled by a pre-strain-controlled correlation length and corresponding characteristic relaxation time that both diverge at a connectivity-controlled critical point, giving rise to weak power law scaling of the complex modulus with frequency and diverging strain fluctuations.

Abstract N° 6

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Model membranes and LLPS mechanics by means of correlative Atomic Force and Confocal microscopies

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Biomechanics is fascinating because the mechanical properties of biological systems can be very diverse due to the high level of heterogeneity and therefore they are highly dependent on the probed lenghtscale and timescale. The viscoelastic behaviour of living cells and tissues is indeed very different from the mechanical response of thin biological membranes or protein liquid-liquid phase separation (LLPS).

In the case of biological membranes, elastic properties are involved in a large number of cellular activites. Membrane mechanics can be assessed at high lateral resolution by means of Atomic Force Microscopy (AFM). We show that the mechanical response of biomimetic model systems such as fluid-gel phase-separated supported lipid bilayers (SLBs) is highly dependent on the size of the AFM tip used as a indenter. In a small tip radius regime (≈ 2 nm) and in the case of fluid phase membranes, we show that the tip can penetrate through the membrane in absence of molecular membrane rupture. For larger indenters (>2 nm) SLBs can be compressed leading to an evaluation of Young's modulus and membrane maximal force before rupture.

Next, in a LLPS frame, we focus on ELF3 plant proteins that have the ability to undergo phase-change in response to environmental signals, such as temperature, enabling these proteins to act as developmental switches. By means of correlative AFM and Confocal microscopies, we evaluate the mechanical response of ELF3 droplets depending on their liquid or gel phase. In the case of liquid phase, we show a force vs indentation linear response which is surface tension dependent, whereas, in the case of gel phase, we observed an elastic response, that leads to the evaluation of the gel Young's modulus.

Abstract N° 7

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Cellular organization in lab-evolved and extant multicellular species obeys a maximum entropy law

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The prevalence of multicellular organisms is due in part to their ability to form complex structures. How cells pack in these structures is a fundamental biophysical issue, underlying their functional properties. However, much remains unknown about how cell packing geometries arise, and how they are affected by random noise during growth - especially absent developmental programs. Here, we quantify the statistics of cellular neighborhoods of two different multicellular eukaryotes: lab-evolved 'snowflake' yeast and the green alga Volvox carteri. We find that despite large differences in cellular organization, the free space associated with individual cells in both organisms closely fits a modified gamma distribution, consistent with maximum entropy predictions originally developed for granular materials. This 'entropic' cellular packing ensures a degree of predictability despite noise, facilitating parent-offspring fidelity even in the absence of developmental regulation. Together with simulations of diverse growth morphologies, these results suggest that gamma-distributed cell neighborhood sizes are a general feature of multicellularity, arising from conserved statistics of cellular packing.

Abstract N° 8

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Structural damping makes insect exoskeleton insensitive to non-sinusoidal deformations

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Muscles act through elastic elements to mediate movement, but these elements can introduce dissipation, which is especially important for energetics during unsteady movements. Recent work has shown that the high power requirements of flapping flight can be reduced by insect exoskeleton, which acts as a structurally damped spring under purely sinusoidal deformation. However, this narrow dynamic regime does not encompass much of real flight behavior. At steady state, some insects beat their wings asymmetrically. Furthermore, insects often transiently modulate wing beat frequency in response to unsteady forces from the environment. It remains unclear whether non-sinusoidal wing strokes incur extra energetic cost via viscous damping or thorax nonlinearity. We used a vibration testing system to measure the mechanical properties of isolated Manduca sexta thoraces under symmetric, asymmetric, and band-limited white noise excitation. We measured a thoracic stiffness of 2954 N/m and peak force of 1738 mN at 25 Hz and physiological peak-to-peak amplitude of 0.92 mm. Power savings and dissipation were indistinguishable between symmetric and asymmetric conditions, demonstrating that no additional energy is required to deform the thorax non-sinusoidally. Under white noise conditions, dissipation was invariant with frequency, which is inconsistent with the response of a viscoelastic material and highlights the importance of frequency-independent damping in the insect exoskeleton. A structurally damped transfer function fits our measured frequency response, indicating that a linear approximation of thorax elasticity is robust across non-sinusoidal deformations. This work demonstrates the potential of structurally damped exoskeletal material to affect motor control by eliminating the velocity-dependent coupling between muscle and appendage associated with viscoelastic connective tissues.

Abstract N° 9

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Mechanical force derived from transcriptional activity can induce local order and frozen motion in interphase chromosome

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Recent experiments reported that dynamics of human interphase chromosome is accelerated by inhibiting transcription. This result is counter-intuitive as transcription involves the elongation process of RNA polymerase which is known to exert force along DNA. To understand this phenomenon, we investigated, using polymer simulations, how mechanical force derived from transcriptional activity affects the dynamic properties of interphase chromosome. In our simulations, a pair of active forces is exerted on chromosome loci in an extensile manner to mimic the effect of transcriptional elongation. We observe that the dynamics of the active polymer chain becomes slower than that of a passive one, which is already in a glassy regime. We demonstrate that the active force gives rise to local crystallization along with frozen motion in a particular range of the force magnitude. By noting the extent of agreement between our results and experimental observations, we suggest the possibility of local order in interphase chromosome induced by transcriptional activity and its contribution to chromatin structure and dynamics.

Abstract N° 10

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Exploring relative advantages of dual vs single dimensionality reduction

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The current experimental advancements in studying complex systems enable the recording of the activity of a large number of constituents of the system, while simultaneously recording the resultant system behavior. Usually, the modeling process begins with dimensionality reduction. Here we explore the dual dimensionality reduction, aiming to identify simultaneously which collective dynamics from the individual constituents are responsible for which reduced dimensional response features. Using linear modeling, we show that the dual dimensionality reduction approach requires significantly fewer data points than the regular methods (i.e., reducing each feature independently, and then identifying relations between the reduced descriptions). However, increasing the number of recorded dimensions is reflected as more noise in the measurements (quantified by the Signal to Noise Ratio - SNR). We numerically investigate the effect of SNR and the number of samples or observations N on single versus dual dimensionality reduction, showing in which regimes and conditions dual reduction is better and when both are the same. Then we outline how we study these results analytically.

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Abstract N° 11

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Robust Altruism by Similarity in Multifaceted Phenotypes

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Altruism remains a difficult phenomenon to predict, and although established theory has supported the existence of altruistic traits for over fifty years, specific mechanisms are controversial. We study the evolution of altruistic behavior under a model where individuals choose to cooperate with each other by comparing a set of always-visible continuous tags, and find the generic maintenance of robust altruism when individuals carry multiple tags. Our model marries existing weak and strong selection results in the literature and addresses challenges to prior model formulations. Our results support the viability of a simple memory-less mechanism for altruism in a well-mixed population by demonstrating robust cooperation buoyed by a regime of weak positive selection that serves as a buffer for strongly selected cycles of high altruism.

Abstract N° 12

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Modeling multi-trophic ecosystems with consumer resource model

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MacArthur's consumer-resource model describes the ecologicaldynamics of a simple two trophic-layer ecosystem where consumerscompete for a pool of resources (e.g herbivores compete for plants).However, real ecosystems often possess multiple trophic layers (e.g.carnivores, herbivores, plants) that all interact to determine ecosystemdynamics. Inspired by this observation, we analyze a generalization ofMacArthur's consumer model with multiple trophic layers using methodsfrom statistical physics. For the three-layer model, we discuss the newproperties of the model, connection with constrained optimization, andpresent a mean-field solution for the multi-trophic layer model using thecavity method. We show that the introduction of additional trophic layersinto an ecosystem can result in both quantitative and qualitativechanges in ecosystem-level properties.

Abstract N° 13

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Kinetics of stator recruitment and release in the bacterial flagellar motor

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The Bacterial Flagellar Motor (BFM) is the molecular machine responsible for the torque generation that produces the rotational movement of flagella in bacteria, allowing them to move through aqueous media. Thanks to its mechano-sensitivity, the motor is capable of adapting to external forces by a system of active association and dissociation of stators, which are ion channels that consume the proton motive force to generate the torque that allows the motor to turn. Despite being one of the most studied nano-machines, the BFM possesses dynamical features that have not yet been fully explained within the framework of statistical physics.

We aim to explain the stochastic behaviour of the recruitment and release processes of stators from the BFM by considering effects such as stator cooperativity and the small size of the system. We use theoretical (lattice gas) and numerical (Monte Carlo simulations) tools to investigate these processes and evaluate the importance of these effects.

Abstract N° 14

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Evolutionary rescue in a fluctuating environment

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Microbes evolve in an environment that is never truly constant. For example, this is the case for a population undergoing antibiotic treatment, when the concentration successively goes below and above the minimum inhibitory concentration. Such fluctuations have a non-trivial effect on evolutionary dynamics, especially when the extinction and environmental time scales are similar. Whereas the impact of deterministic changes has been studied, the impact of stochastic changes is not well understood. To investigate this question, we develop a model in which a population evolves in an environment switching successively between two states: one in which it declines toward extinction, the other in which it grows until reaching its equilibrium size. By combining numerical and analytical approaches, we shed light on two regimes delimited by the mean extinction time in the stressing environment. If the duration of each environmental phase is much shorter than this time, the stochastic switches strongly reduce the persistence time of a population and thus its probability to be rescued by a mutation, compared to deterministic ones. The opposite occurs in the second regime, where environmental time scales are larger than those of extinction in the stressing phase. We derive and validate with simulations analytical predictions of the mean extinction time, the probability of rescue, and the appearance time of a successful mutant. Our work takes an important step towards quantifying evolutionary dynamics in fluctuating environments that could, for example, improve our understanding of the evolution of antimicrobial resistance.

Abstract N° 15

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Impact of finite size effects on the exploration of fitness landscapes by asexual microbial populations

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The exploration of fitness landscapes (genotype vs fitness) by asexual microbial populations is driven by natural selection and genetic drift acting on random mutations. If mutations are rare and selection is strong, the population follows an adaptive walk, so that it can only go uphill in the fitness landscape. However, genetic drift (which arises from finite size effects) impacts this exploration by allowing deleterious mutations to fix in the population. In order to take into account this feature, we drop the strong-selection hypothesis, and study the exploration of rugged fitness landscapes in the weak-mutation regime only. To do so, we use two models, which are respectively based on the Moran model and on the Wright-Fisher model, for the fixation probability of mutations and the resulting dynamics in the fitness landscape. In particular, we show that the mean fitness of the first encountered fitness peak has a non-trivial dependence on the population size. In addition, using multiple fitness landscape models, we show that these finite size effects have a non-trivial dependence on the ruggedness and accessibility of the fitness landscapes.

Abstract N° 16

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Quantification of Myxococcus xanthus Aggregation and Rippling Behaviors with Deep-Learning Approaches

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Myxococcus xanthus bacteria are a model system for understanding pattern formation and collective cell behaviors. When starving, cells aggregate into fruiting bodies to form metabolically inert spores. During predation, cells self-organize into traveling cell-density waves termed ripples. Different issues exist when we are trying to perform analysis.

Both phase-contrast and fluorescence microscopy are used to observe these patterns, but each has its limitations. Phase-contrast images have higher contrast, but the resulting image intensities lose their correlation with cell density. The intensities of fluorescence microscopy images, on the other hand, are well-correlated with cell density, enabling better segmentation of aggregates and better visualization of streaming patterns in between aggregates; however, fluorescence microscopy requires the engineering of cells to express fluorescent proteins and can be phototoxic to cells. To combine the advantages of both imaging methodologies, we develop a generative adversarial network that converts phase-contrast into synthesized fluorescent images. By including an additional histogram-equalized output to the state-of-the-art pix2pixHD algorithm, our model generates accurate images of aggregates and streams, enabling the estimation of aggregate positions and sizes, but with small shifts in their boundaries. Further training on ripple patterns enables accurate estimation of the rippling wavelength. Our

methods are thus applicable to many other phenotypic behaviors and pattern formation studies.

To compare emergent behaviors of different strains, we need to find a low-dimensional set of features that are biologically informative but insensitive to the effects of stochasticity and environmental variability. However, manually selected features often showed more variation between replicate aggregation movies from the same strain than with other strains. Therefore, we built a feature extraction model with both an encoder and a generator. The encoder provides a way to encode M.xanthus images into the low-dimensional feature space. The style-based generator enables us to disentangle aggregate distribution information into the noise dimensions and generate images from low-dimensional feature space. The encoder and generator can work together to evaluate the similarity of different aggregate images and tell the possible genetic relationship in different clusters.

Abstract N° 17

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Deciphering the genome architecture in a minimized bacterial genome

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Probabilistic interactions of RNA polymerase with DNA dictate transcription, the process which evolves the bacterial transcriptome composed of rRNA, tRNA, mRNA, and sRNA. The bounda-ries of the transcriptional events are outlined by the genome architecture defined as the local arrangement of genetic features identified via sequence motifs (promoters, Shine-Dalgarno, gene coding regions, and transcription termination sites, etc.) along the genome. The local ar-rangement of these sequence motifs form operon-like regions, transcriptional units, directly controlling the expression of encoded RNA species. Using structural and bioinformatic analysis of these genetic features, their locations were predicted computationally and integrated to-gether to define the transcriptional units within the minimized genomes of the minimal cell, JCVI-syn3A with 493 genes, and a previous iteration, JCVI-syn1.0 with 868 genes. The computa-tional predictions agree well with the experimental genome architecture identification ob-tained using Oxford Nanopore Technologies which provide long read native RNA sequencing. Comparison between the two genome architectures has resolved the perturbations arising from genome minimization and their potential impact on the cell's phenotype.

Abstract N° 18

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Understanding target selection in high-fidelity Cas9 variants

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The need to limit off-target cutting in DNA-editing applications has led to the engineering of high-fidelity variants of the Cas9 enzyme. One approach hypothesized that the elimination of some non-specific interactions between Cas9 and the DNA substrate could reduce binding affinity and improve target selectivity. However, despite a reduced off-target cleavage rate, no significant difference in binding affinity was observed between wild-type and high-fidelity variants. Therefore, the mechanisms responsible for Cas9 specificity are currently not well understood. Here, we have developed a kinetic model for Cas9 target selection that accounts for RNA-DNA hybridization dynamics and internal conformational dynamics of the Cas9 protein. Our model was parameterized using single-molecule FRET data and high-throughput binding data and predicts cleavage error rates in-line with experimental observations. In addition, we show that the Cas9 specificity enhancement cannot be a result of microstate free-energy perturbations alone, which is consistent with prior theoretical results. Finally, we show that Cas9 conformational dynamics are the most significant contributor to substrate selectivity in PAM-distal mismatched substrates. Our results clarify the mechanisms of Cas9 specificity and suggest new avenues for further Cas9 optimization.

Abstract N° 19

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DNA motors as sensors: toward a new platform for SARS-CoV-2 diagnostics

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The SARS-CoV-2 pandemic has affected over 260 million people worldwide and is still prevalent as new variants continue to emerge. While rapid developments in serological and PCR based assays have increased the diversity and availability of testing for SARS-CoV-2, this is still not sufficient and there still exists a big gap in testing the population. The main problem is that clinical testing of individuals remains uncomfortable, slow, and simply not dynamic enough to prevent the spread of rapidly evolving viral targets. Thus, there is a significant need for rapid, low-cost, at-home testing that is able to identify different strains with high specificity and sensitivity. Such new diagnostic testing is required to help end the current pandemic and avert future ones that will certainly arise in the future. We have developed a platform for viral sensing that integrates a rolling DNA motor that moves micron-size distances. The motor consists of a DNA-coated spherical particle (5 µm diameter) that hybridizes to a surface modified with complementary RNA. The particle moves with speeds of over 1 micron/minute upon addition of ribonuclease H (RNase H), which selectively hydrolyzes hybridized RNA. The motion of the motors can be detected using a conventional smartphone camera equipped with a magnifying lens. To detect the SARS-CoV-2 virus, the DNA motors are modified with virus binding ligands such as aptamers that cross-link the motor to the surface and lead to motor stalling. Therefore, the motors will only stall in presence of SARS-CoV-2 virus particles.

Using motors modified with multivalent aptamers that have high affinity for spike protein, we were able to detect femtomolar range concentrations (104 viral copies/mL) of SARS-CoV-2 WA1 strain as well as the Delta variant in real time and without any sample preparation steps. We demonstrate sensitive and specific detection of SARS-CoV-2 viral particles using DNA-based motors. We show that the motors stall in the presence of a single viral particle and can be readout using a smartphone. Integrated with microfluidic chips enabling smartphone readout, this assay has the potential to enable rapid, low-cost, point-of-care screening for SARS-CoV-2 and other circulating viruses.

Abstract N° 20

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Fundamental Cell Behavior Emerges from Whole Cell Simulations of a Minimal Cell by Integrating Experiment and Theory

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An overarching goal of molecular biology is to explain the basic processes of life in terms of the laws of physics and chemistry. Just as physics had studied the hydrogen atom as its simplest system to understand more complex atoms and molecules, a simplest, minimal cell would be the ideal platform to study the fundamental behavior of cellular life and expand the principles to more complex cells. JCVI-syn3A is a genetically minimal cell consisting of 493 genes, 452 of which code for proteins, on a singular 543 kbp circular chromosome. The cell cannot be described by a simple small set of equations like the hydrogen atom, but it can be simulated as a whole-cell model of all the chemical reactions inside the cell. Whole-cell modeling requires the combination of multiple simulation methods and the results of many experimental techniques to inform dynamics and cell architecture. We constructed a whole-cell kinetic model of Syn3A including the complete metabolic network, cell growth, and genetic information processing reactions for DNA replication initiation and elongation, transcription of all 493 genes, and translation and degradation of all 452 mRNA. Parameters were obtained from single molecule experiments and kinetic parameter databases such as BRENDA and TECRdb. A spatial model of Syn3A used cell architecture constructed from cryo-electron tomograms. From the mRNA diffusion and degradation reactions occurring at membrane-bound degradosomes, we predict the distribution of mRNA half-lives. A well-stirred version of the model simulated for complete cell cycles includes multiple DNA replication initiation and elongation events per cell cycle, which qualitatively agrees with qPCR experiments and was observed to help maintain homeostasis in metabolism. These novel simulations track the exact time-dependent ATP use of each reaction in the simulation.

Abstract N° 21

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Bacteria use a two-stage mechanosensing process to initiate biofilms

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The attachment of bacteria onto a surface, consequent signaling, and the accumulation and growth of the surface-bound bacterial population are key initial steps in the formation of pathogenic biofilms – yet whether, and how, this is impacted by the mechanics of the surface is not known. We use ultrathin and thick hydrogels to create stiff and soft composite materials, respectively, with the same surface chemistry. Using quantitative microscopy, we find that the accumulation, motility, growth and cyclic-di-GMP signaling of the opportunistic human pathogen Pseudomonas aeruginosa respond to surface stiffness. Using finite-element modeling combined with experiments, we find that adhesion to stiffer surfaces results in greater changes in mechanical tension in the bacterial envelope than does adhesion to softer surfaces with identical surface chemistry. The cell-surface-exposed protein PilY1 acts as a mechanosensor, resulting in higher cyclic-di-GMP signaling, lower motility, and greater accumulation on stiffer surfaces. Subsequently, PilY1 makes the biofilm lag phase longer for bacteria attached to stiffer surfaces. In the exponential phase that follows the biofilm lag phase, populations of bacteria with functional pilus retraction motors PilT grow faster on softer surfaces. This demonstrates a two-stage mechanoresponse in early biofilm formation.

Abstract N° 22

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Force-sensing through ion channels

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Mechanosensation is essential for cellular survival, and was already identified by Aristotle as one of the five fundamental senses. In bacteria, mechanosensitive ion channels have long been known to transduce mechanical force into an electrochemical response. But the molecular basis for mechanosensation in mammals is only beginning to be unraveled. Over the past few years it has been established that Piezo ion channels underlie mechanosensation in a wide range of mammalian cells. We show here that membrane mechanics provides a unifying physical framework for understanding mechanosensation in bacteria and complex multicellular organisms. Applying this framework to mechanosensation in mammals we find that, at small membrane tension, Piezo tends to locally curve the cell membrane into a minimal surface. The unique mathematical properties of this minimal surface amplify the sensitivity of Piezo to changes in membrane tension, rendering it exquisitely responsive. We assert that the shape of Piezo is an elegant example of molecular form evolved to optimize a specific function, in this case tension sensitivity.

Abstract N° 23

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Signaling through formation of a pre-wet surface density

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Membrane-bound receptors sense ligands at the plasma membrane, and must amplify and communicate the signals they receive into the three-dimensional cytoplasm. Proteins involved in these signaling cascades often form "signaling clusters" upon activation: structures rich in many of the 3D components required for downstream signaling, as well as specific lipids and membrane bound proteins. Recent work has suggested that liquid-liquid phase separation underpins many signaling clusters. Here we develop a theory of these phenomena through Monte-Carlo simulations and a minimal Landau theory. Our model suggests that signaling clusters are best described as prewet, displaying surface-only phase coexistence outside of the ordinary membrane and bulk protein coexistence regions[1]. Prewet phases confer the ability to 'integrate' the state of the bulk and membrane concurrently. We explore the ramifications of our model wherein signal initially received at many receptors is integrated and transduced by the dynamics of the prewetting transition.

Abstract N° 24

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Extracting and exploiting multiple temporal features of odor signals to navigate complex plumes

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Yale University

The unpredictable nature of turbulence makes navigating realistic odor plumes highly difficult. Yet even creatures as simple as fruit flies can excel at this task. It is known that during odor plume navigation, walking Drosophila melanogaster bias their motion upwind in response to both the frequency of their encounters with the odor (Demir et al., 2020) , and the fraction of time the signal is above a detection threshold (Alvarez-Salvado et al., 2018) . In my talk, I will describe how sensing both features is necessary for robust navigation of diverse turbulent odor plumes and how these features should be combined into a navigation model. I will also explain how known models of odor transduction in the Drosophila olfactory circuit are capable of responding to the implicated temporal features and can be incorporated into simple navigational models. Finally, I will discuss how recent behavioral studies of fruit flies in environments with temporally diverse odor signals reveal a rich and varying dynamical control of orientation, influenced by several temporal aspects of the perceived signal.

Abstract N° 25

Eric Mazur Harvard University, USA

The PoLS-Teacher Network: High-school Teachers and the Future of Our Discipline

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Abstract N° 26

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Research-Practice Partnerships to Improve Physics Education

Teacher-Initiated Projects (TIPs) are research-practice partnerships (RPPs) between physics teachers and physics education researchers with the aim to improve physics education, in the classroom. The TIPs are a part of the broader Physics of Living Systems Teacher (PoLS-T) Network and were established to address physics teacher concerns, namely, the lack of resources to implement evidence-based improvements in their physics classroom and physics teacher isolation. The focus of this talk will be on the initial cohort of ten TIP collaborators and the fostering RPPs in an authentic and mutually beneficial way, while also establishing a community of practice to counter physics teacher isolation. Further, TIPs provide an example of how collaborative partnerships that center teacher expertise and leverage researcher resources can improve student physics education.

Abstract N° 27

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The Road Less Traveled- Social Emotional Learning in Physics

Social Emotional Learning better known as SEL in the education community is a concept that has really come into the limelight in recent years. There have been studies linking social emotional learning to better academic performance as well as better living as the students become adults. Yet most of the studies have concentrated on early childhood. Here I try to present ways in which SEL is as important to the physics community as it is to any other and also show very briefly key components needed for this to be true. I will discuss some perceptions students may have about physics teachers and also about minorities in physics and ways to address this through SEL.

Abstract N° 28

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Training the mind to think: The importance of mindset in physics

Even the most basic studies in Physics exposes students to new and exciting levels of understanding the world around them; It also offers a unique opportunity to train the mind to think in new ways. Physics provides a framework for students to improve their problem solving, diligence, perseverance, and data analysis skills. While raw IQ can have a meaningful effect on students' success in Physics, character and perspective play an outsized role on student achievement. Specifically, the aggregate mindset of the teacher-student relationship has been indicated as a strong predictor of student engagement and academic performance. This growth mindset shifts the emphasis from a potentially arbitrary obsession with results towards an appreciation for the obstacles of learning and, all the while instilling the value of failure along the path to understanding. This session will focus on the role of mindset in student outcomes, highlighting the discussions from the PoLS-T Network talk series sessions.

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Abstract N° 29

Jolene Johnson Osseo Public Schools, North Hennepin Community College, University of Wisconsin River Falls, USA

Addressing the lack of diversity in physics-Equity issues and solutions in high school physics education

It is well documented that in the US (and many other countries) that the physics community suffers from lack of diversity. In this US at least some of the lack of diversity in physics career paths begins with equity issues in high school physics classes. I will present data about variations in access to quality physics courses due to geography and socioeconomic status. Additionally I will present data on the equity issues in high school physics by race and gender. The PoLS-T network has been focused on exposing and seeking solutions to these equity issues through many talks and programs. I will describe some of the teacher lead solutions including STEP-UP (which encourages young women in physics), diversity lessons including from the Underrepresentation Project, and talks and ideas presented at PoLS-T network.

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Abstract N° 30

Fabien Kunis Sofia University St. Kliment Ohridski and Boyn Penev High School, Sofia, Bulgaria, Bulgaria

Think globally, act locally

In this talk I will talk about implementing in Bulgaria the activities of the PoLS-T network. PoLS-T Network is an international network whose primary goal is to identify the major problems in education and potential solutions at the local and global level. Many of the problems discussed can be called global, i.e. these problems occur in one form or another in different countries. I would like to share my experience as a teacher in Bulgaria. Participating in the network, I was able to learn new methodologies and solutions to various problems in education. But every education system is unique in itself. That is why implementing global solutions is a challenge. In this talk, I share my experience of implementing the Peer Instruction and JIGSAW methodologies. I share about the initiatives that the Faculty of Physics of Sofia University St. Kliment Ohridski and the Union of Physicists in Bulgaria have taken to increase the interest of students in the teaching profession. I conclude with my experience in promoting citizen science projects through my participation in CERN CMS Masterclass.

Abstract N° 31

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Creating Community Scientists

Mollie Thurman (1)

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BioBus is a science education nonprofit that specializes in bringing research tools and scientists to communities across New York and New England. With our mobile microscopy laboratories and team of scientists, we have worked with over 300,000 students since 2008. Many of the communities that we work with are directly adjacent to academic institutions and yet most of our students have never had an opportunity to meet a scientist or visit an active research laboratory.

This session will focus on the BioBus Community Scientist Fellowship Program currently being run in partnership with Harvard University's Department of Molecular and Cellular Biology. Through this program, graduate students and postdocs receive training and feedback from seasoned BioBus scientists and work with students from public schools across New England. As scientists improve their outreach and communication skills, they are encouraged to incorporate their own research topics to create unique, informal science activities and curricula.

This model for training scientists and generating curriculum is highly adaptive and responds to many of the challenges that are faced by academic departments who are interested in doing outreach, but don't know where to start.

Abstract N° 32

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Modelling cell motility on asymmetric nanotopography

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Asymmetric nanotopography with sub-cellular dimensions has recently demonstrated the ability to provide a unidirectional bias in cell migration. The details of this guidance depend on the type of cell studied and the design of the nanotopography. This behavior is not yet well understood, so there is a need for a predictive description of cell migration on such nanotopography that captures both the initiation of migration, and the way cell migration evolves. Here, we employ a three-dimensional, physics-based model to study cell guidance on asymmetric nanosawteeth. In agreement with experimental data, our model predicts that asymmetric sawteeth lead to spontaneous motion. Our model demonstrates that nanosawteeth induce a unidirectional bias in guidance direction that is dependent upon actin polymerization rate and sawtooth dimensions. Motivated by this model, a new analysis of previously reported experimental data indicates that the degree of guidance by nanosawteeth increases with the cell velocity.

Abstract N° 33

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Controlling antagonist responses with a single activator

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Cell migration is a particularly important process in cancer development, as it is required for the formation of metastases, which are themselves responsible for 90% of cancer-related deaths. Single cell migration is controlled by a fine balance between contraction and extension of its cytoskeleton, especially actin and myosin. In mesenchymal migration, cytoskeleton extension takes place at the cell front by the activation of the protein Rac, which induces actin polymerization and therefore formation of membrane protrusions adhering to the substrate and propelling the cell forward. At the back, the cell contracts thanks to actin fibers formation and myosin activation, two processes known to be controlled by the activation of the protein RhoA.

By building an optogenetic tool allowing precise spatial and temporal control of RhoA activity, we challenge this simplistic view of the role of RhoA. Indeed, activating RhoA locally in a single cell sometimes leads to contraction as expected, but other times to protrusion, thus polarizing the cell in two diametrically opposite directions with the same molecule. How can the same protein induce two opposite phenotypes, and therefore two opposite polarizations, in the same cells?

We show that the two phenotypes are triggered by different signaling cascades, which arise above all from the dynamics of temporal activation of RhoA at the membrane, and not from the proteome of the cell. This temporal activation of RhoA is due to both to the basal

level of our activator and the dynamic of its activation: the system falls in two opposite states (contraction or protrusion) depending on the amount of activated myosin before the activation, a system that challenges the classical view of the contractility pathway.

Abstract N° 34

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Electrical characterization of single bacterial cells

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Bacterial electrodynamics at the single cell level is still largely unexplored. Due to constraints in cell size and the presence of cell wall, voltage is inaccessible to direct electrophysiology techniques such as patch clamp. This limits our knowledge of the dynamics of the proton motive force (PMF), the electro-chemical potential across the inner membrane that regulates a wide range of physiological processes such as ATP synthesis or rotation of the bacterial flagellar motor (BFM). Because the speed of the BFM is linearly proportional to the PMF, it can serve as local PMF sensor on the membrane [1]. Previous studies have shown that illumination of bacteria expressing a light-driven proton pump proteorhodopsin (PR) generates a PMF capable to power the BFM [2,3]. Here, we perform high resolution measurements of BFM speed response to controlled PMF changes upon multiple and short illumination cycles in Escherichia coli expressing PR. Based on previous work where components of electrical circuits serve as analogy to relevant bacterial elements involved in PMF dynamics [2], we model the BFM as an electro-mechanical voltmeter whose rotation and proton flux are tightly coupled. Analytically solving the circuit and performing a global fit of the model to our experimental traces, we show that each cellular component can be quantified, yielding dynamical access to the electrical activity of the cell.

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Abstract N° 35

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On the adhesion-velocity relation and multistability of the motile state of MDA MB 231 cells on fibronectin lanes

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Migration of eukaryotic cells is a fundamental process for embryonic development, wound healing, immune responses, and tumour metastasis.

A universal observation is the well-known biphasic adhesion-velocity relation.

There is, however, little quantitative understanding of how adhesion and intracellular forces control cell velocity.

We study the motion of MDA-MB-231 cells on microlanes with fields of alternating Fibronectin densities to address this topic and derive a mathematical model from the leading-edge force balance and the force-dependent polymerization rate.

It reproduces quantitatively our measured adhesion-velocity relation. All motion-related forces are controlled by adhesion and velocity, which allows motion even with higher Fibronectin density at the rear than at the front.

At transitions between different Fibronectin densities, steady motion is perturbed which changes the front and rear velocity.

We then discuss the role of the biphasic relation between retrograde flow velocity and friction force for transitions of motile states.

Abstract N° 36

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Genomic physics: biophysical modeling confronted to genomic data

Jean-Charles Walter

Laboratoire Charles Coulomb (L2C)

The last decade has seen a spectacular development of genomic physics both in bacteria and eukaryotes. This is due in large part to the development of experimental techniques, which allow to probe (i) the motion of nucleic acids (chromosomal DNA and RNA) in vivo by means of microscopy techniques and (ii) the interactions of these nucleic acids with other actors of the cell (like proteins and molecular motors) by means of high-throughput sequencing techniques. I will focus on three different processes : (i) the modeling of translation with a ballistic model describing the motion of ribosomes on messenger RNA, (ii) the physical mechanism of liquid-liquid phase separation, which is found to be universally spread in the cells to increase, locally and transiently, the concentration of a molecular actor (proteins, RNA etc.). We illustrate this process with the example of the bacterial DNA segregation where liquid-liquid phase separation is used to increase the local concentration in order to catalyze ATP hydrolysis. Finally (iii) I give a general framework on how polymer physics can help to infer the organization of DNA in vivo from microscopy and high-throughput sequencing experiments and apply it to the modeling of bacterial DNA in Escherichia coli and to the epigenetic regulation via polycomb domains in Drosophila.

Abstract N° 37

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3D diffusion in live Escherichia coli cells.

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The bacterial cytoplasm is a crowded and polydisperse environment which leads to interesting anomalous diffusion of intracellular macromolecules. Using a combination of biplane microscopy and single particle tracking, we reconstruct the 3D motion of Genetically Encoded Multimeric nanoparticles (GEMs) inside Escherichia coli cells to study the rheology of the bacterial cytoplasm. We use GEMs ranging in size from 20 to 50 nm, similar in scale to ribosomes and other complexes in the cell. The motion of larger particles is confined to cylindrical shells around the nucleoid, with several diffusive regimes. Additionally, we study exponential and stationary phase cells to understand the effects of metabolic state on diffusion. We observe that the nucleoid in exponentially growing cells contains ~50-nm sized compartments that cage diffusing particles. Entrance into stationary phase involves rearrangements of the nucleoid that allow diffusion of particles of multiple size scales.

Abstract N° 38

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Modeling plant root circumnutation using cellular simulation

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Circumnutation-the helical motion of a growing plant root tip-is hypothesized to benefit plants by increasing the probability of penetration through heterogeneities in soil [Taylor et al, PNAS 2021], but it is not well understood how this behavior is initiated and maintained in the root. Modeling the root computationally can provide insight into the mechanics of this phenomena in ways not easily achievable with genetics alone. Previously, researchers have attempted to theoretically study circumnutation by modeling the root as a series of cylindrical segments and demonstrated its emergence through gravitropism [Ilya Loshchilov, PLOS 2021]. To understand how circumnutation develops from cell level behaviors, we developed a numerical simulation using a DEM (Discrete Element Modeling) approach to model plant roots. Since the biological root is separated into the elongation, transition, and meristematic zones, as well as the root cap, the simulated root is treated as a growing array of cells which divide close to the tip and begin to elongate in the transition zone; these are each represented as a particle with a fixed width and variable length to allow for elongation during osmotic expansion. This simulation was tested for a range of parameters intended to capture laboratory experiments of rice roots grown in gel [Taylor et al, PNAS 2021], with a maximum of approximately four thousand cells in the simulation. Using this simulation, we find an algorithm to model circumnutation as a behavior created from nearest neighbor coupling of cells to elongation rates, allowing for a delayed onset of coordination. We compare the simulation to both wild-type circumnutating and mutant non-circumnutating biological roots. We also analyze the curvatures over time of both the biological experiment

and simulation and note that the frequency of circumnutation decreases at 24 hours in rice roots.

Abstract N° 39

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Membrane-binding domain affects the diffusion of the RNA degradosome in Escherichia coli

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The RNA degradosome is the major protein complex responsible for mRNA turnover in many bacterial species. In Escherichia coli the RNA degradosome is spatially separated from the sites of transcription and translation via an amphipathic motif in RNase E, called a membrane-targeting sequence (MTS). MTS associates with the inner membrane. However, it is uncertain how the RNA degradosome's diffusion (and thus mRNA lifetime) is affected by the MTS. Here we use super-resolution single-particle tracking to study the effect of the MTS on the degradosome diffusion by two different perturbations. The first perturbation is removing MTS from RNase E. The second is to replace MTS with a transmembrane motif. We found that cytoplasmic RNase E without MTS diffuses more quickly than RNase E associated to the inner membrane via the MTS. Additionally, molecules with MTS diffuse more quickly than molecules with the transmembrane motif.

Abstract N° 40

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Programming collective cell migration with bioelectric cues

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The wonderful diversity and capabilities of multicellular systems are possible, in part, because of beautifully coordinated processes such as the collective migration of myriad interacting cells. As collective cell migration regulates processes spanning morphogenesis, healing, and disease, imagine the potential of being able to externally direct the collective motion of thousands of cells, akin to how a sheepdog herds sheep. Such control of cellular locomotion would allow us to interact with collective cellular processes in new ways, such as

reprogramming tissue form to produce new structures or accelerate healing. However, realizing this potential requires both unique collective analysis tools, and a 'sheepdog' stimulus capable of interactively controlling collective cell migration. For analysis, my group has developed a variety of biophysical and machine learning approaches to extract biophysical parameters from cellular collectives. As for the 'sheepdog,' we are working to develop electrotaxis—directed cell migration along direct-current (DC) electrical gradients—into a tool to reprogram collective cell behavior. DC fields arise naturally in multicellular systems via ion gradient disruption during development, disease, and injury, such as in skin wounds. Such ionic fields are thought to induce electrophoresis of charged membrane receptors and to deform ion channels, ultimately triggering intracellular signaling to guide migration. To both better understand and harness electrotaxis, we built a new bioreactor called SCHEEPDOG that can spatiotemporally program 2D electric fields in mm-scale tissues. This approach recently revealed both that cells time-average electrical cues on the minute-scale, and that tissues retain a long-term memory of having been stimulated, reflected in a sustained change to collective properties post-stimulation. Interestingly, we have also found that attempting to program collective cell migration can set up a 'tug-of-war' within the tissue between the external command and strong, pre-existing collective behaviors within the tissue. This conflict can be mitigated by transiently weakening cell-cell coordination, making the tissue more responsive to external control, allowing us electrically increase in vitro 'healing' speeds in scratched skin layers by 2-fold. Finally, we have been exploring the stimulation of 3D structures such as hollow organoids, where stimulation induces rapid hydrostatic swelling regulated by ion channels.

Abstract N° 41

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Impact of glycosylation on uptake of receptors by endocytosis

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Clathrin-mediated endocytosis is a fundamental cellular uptake process that is essential for nutrient import, signaling, and recycling of membrane proteins. Many membrane proteins that are internalized by endocytosis are also glycosylated on their extracellular domains. Some of these glycosylated proteins are constituents of the glycocalyx, a cellular structure that provides a physical barrier to exclude pathogens from cells and maintains membrane hydration. Despite the essential role of membrane traffic in maintaining the glycocalyx, little is known about the impact of glycosylation on the uptake of membrane proteins. To study this relationship, we replaced the ectodomain of the transferrin receptor, a well-studied membrane protein that undergoes endocytosis, with a highly glycosylated domain from MUC1, a constituent of the glycocalyx. When we overexpressed this model receptor in mammalian epithelial cells, we found that endocytic uptake of the glycosylated receptor was substantially reduced in comparison to a non-glycosylated control protein. We observed similar results when we inhibited glycosylation of the model receptor by treating cells with benzyl-α-GalNAc. Compared to the glycosylated model receptors, the glycosylation deficient receptors showed increased partitioning into clathrin-coated pits. We hypothesize that the bulky ectodomain of the glycosylated receptors presents a steric limitation, which decreases the effective capacity of clathrin-coated pits to internalize the receptor. Specifically, fewer copies of the bulky, glycosylated model receptor can be accommodated within clathrin-coated pits. These results suggest a physical role for glycosylation in maintaining the glycocalyx. In particular, glycosylation may provide a novel physical signal for keeping receptors on the plasma membrane, rather than allowing them to be recycled and degraded. Therefore, our result may have important implications for regulation of the glycocalyx in its many diverse rolls, from the prevention of pathogen entry to modulating the behavior of cells during inflammation and carcinogenesis.

Abstract N° 42

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Microfluidic flow of vesicle prototissues as a model for cellular tissues

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To circumvent the barriers to the comprehension of cellular behavior and its contribution to biological processes, the emergence of biomimetic tissue engineering has proved to be a valuable ally. Following this bottom-up approach, it has become possible to design model tissues. By isolating specific physical or chemical mechanisms, the complexity of biological processes such as morphogenesis can be broken down. Here, we focus on the mechanical characterization of such prototissues, specifically in flow conditions, based on a microfluidic approach.

Starting with the synthesis of custom Giant Unilamellar Vesicles (GUV), we proceed to assemble them through Streptavidin-Biotin based adhesion to obtain spheroidal prototissues. These prototissues show tunable vesicle-vesicle adhesion and size. Moreover, the latter can be comparable to the one of embryonic tissues. The assembled prototissues are then mechanically probed in a "pipette-aspiration" inspired microfluidic chip, coupled with epifluorescence microscopy, under controlled flow and pressure conditions.

We quantify the rheological response of the prototissue to a creep experiment and we describe the viscoelastic behavior observed with a modified Maxwell model. Moreover, flow velocimetry and fluorescence quantification provides the monitoring of phenomena happening at the meso scale such as tissue reorganization and vesicle-vesicle rearrangement.

This rheological tool allows the simultaneous study at two different length scales making a bridge between the global viscoelastic response of the whole prototissue and the adhesion properties between the individual vesicles. Designed for vesicle prototissue assays, it can be translated to the study of embryonic tissues, thus allowing a deeper comprehension by comparing the two tissular systems. This parallel can help giving valuable insights into the complex phenomena at the tissue scale observed during morphogenesis.

Abstract N° 43

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Mixing in microcirculation

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Rapid mixing is of essential importance in many microfluidic applications such as during

homogenization of solutions of reagents used in chemical reactions, sequencing or synthesis of nucleic acids or protein crystallization. However, it is notoriously difficult to mix fluids in micro- channels at low Reynolds number conditions of laminar flow where molecular diffusion is usually insufficient to mix fluids across the channel size on the time scale of its usual residence time, especially in solutions containing macromolecules that have diffusion coefficients one or two orders of magnitude lower than that of ordinary small molecules. This problem is crucial in blood flow since b lood is known to transport oxygen by red blood cells (RBCs) as well as proteins, glucose, mineral ions and hormones by the plasma, its liquid suspending phase. Macromolecules and other moieties are constantly exchanged between blood and the endothelial cell. A good mixing is therefore essential for the organism. In parallel, during the past few decades, synthetic micro- and nano-particles have also been transported by the blood stream used as drug, gene and/or imaging delivery vehicles [1] and there mixing and availability near the vessels walls is a prerequisite to their function.

We propose in this work to discuss the importance of blood being a concentrated suspension of RBCs in the mixing process that happens everyday in the body. By the use of squared-cross- sectional microchannels, we investigate the mixing of fluorescent macromolecules in a suspension of RBCs of different volume fractions and flow rates around the physiological value. We show that mixing process depends not only on the volume fraction of RBCs but also strongly on the flow rate. Mixing improves

for higher flow rates for fix RBC volume fraction values. Besides, an optimum RBC volume fraction has been found to maximize the mixing process. Our results shown that RBCs are not only essential bodies for the transport of oxygen, but are also fundamental entities acting directly on the improvement of macromolecule mixing present on their liquid suspending phase. Finally, we proposed our system as a model for other problems associated to microfluidics applications, where the mixing process is essential.

Abstract N° 44

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Bacterial Growth Mechanisms and Their Role in Cell Size Homeostasis

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Growth is a fundamental feature of living organisms which is determined by the rate of biochemical reactions in the cell, their efficiency, and their collective organization. The aim of this work is to uncover mechanisms that control bacterial growth rate and understand how they contribute to cell size. We applied a linear regression analysis to multiple simultaneously measured cellular properties to separate indirect correlations from key dependencies. This new regression analysis is not biased by preexisting models or proposed biological mechanisms, and its results reveal three features: 1- the size fraction a cell receives from its mother is independent of other variables, 2- the growth rate of the cell depends on the growth rate of its mother and on the size fraction it received during division, and 3- the relative growth during each cycle is regulated by its initial size, and the growth rate of the same cell cycle. The second feature was unpredicted, and therefore we further investigated it. This result suggests a mechanism of growth rate regulation that is inverse proportional to the size fraction the cell receives from its mother and thus acts to make up for the size difference between two sister cells. Note that, it is not the absolute size of the daughter, but rather the relative size compared to the mother, that affects the growth rate. Most resources in the cell are expected to be distributed between the cells with equal

concentration with some random noise that cannot lead to a measured bias; DNA is the only resource that is distributed between two cells in equal amounts. Thus, a cell with a smaller size fraction will have a higher concentration of DNA. We hypothesize that the higher concentration of DNA leads to higher production rate of proteins in the cell and as a result a larger growth rate. Such mechanism can provide a feedback source to offset size difference between cells, which is fundamental for size homeostasis. To test this hypothesis, we developed an experimental procedure in which cells were made to filament with and without DNA replication, while growing in microfluidic traps. Our measurements of the growth rate in both scenarios reveal that while cells filament in both cases, when DNA is replicated, the cells growth rate remains constant, while during DNA replication arrest, the growth rate slows down significantly. Our results show that the growth rate of the cell in all cases is linearly proportional to the DNA concentration.

Abstract N° 45

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Towards Simulating Eukaryotic Cells from the Fundamental Physico-Chemical Principles

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One of the key unsolved challenges at the interface of physical and life sciences is to formulate comprehensive computational modeling of cells of higher organisms that is based on microscopic molecular principles of chemistry and physics. Towards addressing this problem, we have developed a unique reactive mechanochemical force-field and software, called MEDYAN (Mechanochemical Dynamics of Active Networks: http://medyan.org). MEDYAN integrates dynamics of multiple mutually interacting phases: 1) a spatially resolved solution phase is treated using a reaction-diffusion master equation; 2) a polymeric gel phase is both chemically reactive and also undergoes complex mechanical deformations; 3) flexible membrane boundaries interact mechanically and chemically with both solution and gel phases. I will demonstrate our recent progress in simulating multi-micron scale cytosolic/cytoskeletal dynamics at 1000 seconds timescale, and also highlight the outstanding challenges in bringing about the capability for routine molecular modeling of eukaryotic cells.

Abstract N° 46

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The Muscular Multitool: Biomechanics of the Elephant Trunk from Macro to Micro

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In 1985, biologist Bill Kier stated that since those muscles are mostly made of water, muscular hydrostats should be incompressible and satisfy volume conservation. It is known that tongues and tentacles obey volume conservation and elongate in a homogenous way due to homogeneities of the internal muscle volume of the structures. The elephant trunk shows important differences between tongues and tentacles. On the outside, the ventral trunk is covered in wrinkles, whereas the dorsal portion has thick folds dotted with vibrissal hairs. On the inside, the cross-section has two nostrils imposing negative space to the three different muscle groups inside the trunk: longitudinal, radial, and oblique groups. In this experimental study, we studied the elongation and reaching biomechanics of the largest terrestrial hydrostat, the elephant. We were surprised to see that during reaching tasks, the elephant asymmetric elongation. The goal of this experimental study is to understand the causes of this asymmetry. At Zoo Atlanta, we perform high-speed videography and tracking of the skin of the trunk. The trunk reaches in a telescoping pattern, from tip to base. The trunk only elongates the larger cross-section at the base when necessary, helping the elephant to conserve energy. The elephant elongates by 25%, but also experiences asymmetry in the dorsal-ventral regions elongating by 25-35% more in certain dorsal sections than the ventral sections. This asymmetry is unique to the elephant, and we hypothesize that it is to the unique structures of the elephant's muscle and skin. We look to understand the morphology and composition of the elephants' muscles and skin that allow this asymmetry and are looking to create new bio-inspired mimics of the elephant trunk that factor the arrangement of collagen fibers at the micrometer scale.

Abstract N° 47

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Quantification of Chaos in Rapidly Paced Cardiac Tissue

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Cardiac tissue is modeled as a nonlinear system that can exhibit chaotic dynamics. However, little experimental evidence has been collected to verify these dynamics. This study aims to both quantify and qualify the chaotic nature of cardiac tissue from a cardiac system's electrical response to a range of periodic forcing signals. This included high frequencies that emulate arrhythmias. Tissue response was recorded as a voltage signal from a single cell within a frog ventricle. The resulting action potential durations (APDs) from the signal were measured as a time series for analysis with Lyapunov exponents. Results have shown to give negative Lyapunov exponents for forcing frequencies around APD bifurcation events and positive exponents during the complex response. In addition, stable period-three orbits and several unstable periodic orbits of APDs were identified during the complex response. This further suggests that the tissue response is not random, but in fact chaotic; informing the development of methods to control and terminate arrhythmias.

Abstract N° 48

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Thermodynamics of Animal Locomotion

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Muscles are biological actuators extensively studied in the frame of Hill's classic empirical model as isolated biomechanical entities, which hardly applies to a living organism subjected to physiological and environmental constraints. In this presentation we elucidate the overarching principle of a living muscle action for locomotion, considering it from the thermodynamic viewpoint as an assembly of actuators (muscle units) connected in parallel, operating via chemical-to-mechanical energy conversion under mixed (potential and flux) boundary conditions. Introducing the energy cost of effort as the generalization of the well-known oxygen cost of transport in the frame of our compact locally linear nonequilibrium thermodynamics model, we analyze oxygen consumption measurement data from a documented experiment on energy cost management and optimization by horses moving at three different gaits. Horses adapt to a particular gait by mobilizing a nearly constant number of muscle units minimizing waste production per unit distance covered; this number significantly changes during transition between gaits. The mechanical function of the animal is therefore determined both by its own thermodynamic characteristics and by the metabolic operating point of the locomotor system.

Abstract N° 49

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Measuring animal behavior in social contexts using deep learning approaches

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Understanding animal behavior in a social context requires tracking and quantifying their behaviors as they interact with each other. Recently, there has been notable progress in developing deep learning algorithms and software to track multiple animals in social paradigms. Some of these algorithms only track animal identities, which has been shown to be robust. Other algorithms detect key points on the animals' bodies to estimate poses, which can be reasonably successful when the animals are some distance apart or have brief close contact with each other. However, these algorithms exhibit poor tracking accuracy when the animals spend

more time with each other, performing key social behaviors such as huddling, mutual-grooming, and mating. To help improve the tracking of animals within social contexts, when the animals are close for long periods, we implemented a pipeline that combines multiple deep-learning-based tracking methods to obtain detailed and high-accuracy postural trajectories of multiple animals. Tested on a data set of prairie voles - a model organism for the study of social interactions - our pipeline robustly maintains animal identities and increases the accuracy of the posture tracking over applying convolutional neural network methods by themselves. With this improved tracking accuracy, we can build a better representation of the behaviors of animals in a social context, isolating behaviors that are key for understanding the dynamics of social interactions.

Abstract N° 50

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Socially isolated bumblebees have perturbed individual and social behaviors compared to socially reared bumblebees

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Bumblebees are eusocial insects that must correctly give and receive social cues to function as a group. Newly eclosed bumblebees go through a nine-day period of rapid brain development as they take on tasks within the hive. To probe how social experience during this time affects later behavior we removed newly eclosed bees from the colony and isolated them for nine days. We observed the behavior of age-matched bees from extra-hive isolated and grouped rearing conditions and hive reared bees. We used SLEAP to track posture and quantified the behavioral repertoire of bees in both individual and paired behavioral trials. We find broad behavior differences between isolated bees and their socially reared counterparts. We also find that paired isolated bees paired have significant changes in behavior at close distances to their partner that socially reared bees lack.

Abstract N° 51

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Alternation emerges as a multi-modal strategy for turbulent odor navigation

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Foraging mammals often pause to sniff in the air preceded by rearing on their hind legs or raising their head. Interestingly, this behavior emerges spontaneously during olfactory search in presence of airflow, suggesting that alternation may serve an important role during turbulent plume-tracking [1]. To test this hypothesis, we combine fully-resolved numerical simulations of turbulent odor transport and Bellman optimization methods for decision-making under partial observability (POMDP). We show that an agent trained to minimize search time in a realistic odor plume exhibits extensive alternation between sniffing the air vs the ground, together with the characteristic cast-and-surge behavior commonly observed in flying insects. Alternation is tightly linked with casting and occurs more frequently when the agent is far downwind of the source, where the likelihood of detecting airborne cues is higher relative to cues close to the ground and uncertainty dominates. Casting and alternation emerge as complementary tools for effective exploration when cues are sparse. Fluid dynamics explains the alternating behavior: in fact when the environment is turbulent odor cues travel faster and further in the air than at the ground were they are slowed down by the boundary layer. We develop a model based on marginal value theory to capture the interplay between casting, surging and alternation. More generally, we show how multiple sensorimotor modalities can be fruitfully integrated during complex goal-directed behavior.

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Abstract N° 52

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Modeling Learning behavior in C.elegans

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The roundworm C. elegans learns from its experiences. Worms on a thermal gradient perform thermotaxis to or away from the conditioned temperature, depending on its thermal history and the starvation duration during the conditioning phase. To quantify this behavior, we developed a novel assay that tracks single worms—each experiencing a spatiotemporal thermal gradient with thermal precision ±0.01C—in a small (2.8ul) droplets of buffer, arrayed on hydrophobic-printed microscope slides. CCD cameras track individual worms simultaneously in many droplets for many hours and at high temporal resolution, each droplet set at 20C midpoint with a thermal gradient of 0.5C/cm. A worm's thermal preference is summarized as a thermotaxis index. Initially, worms reared at 25C and 15C exhibit thermophilic or cryophilic tendencies, respectively, and starvation during conditioning or in the droplet reverses these tendencies. This reversal indicates multiple dynamic processes for learning that operate on different time scales. We build a predictive model with multiple time scales and utilize mutants to detangle the various learning processes. The model predicts the behavior under various training conditions and genetic perturbations.

Abstract N° 53

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Body oscillations couple with wing flapping to reduce the aerodynamic power in wild silkmoths

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Quasi-steady models of insect flight often prescribe constant body dynamics during a wingstroke. However, many insects experience large fluctuations in body kinematics even during a single wingstroke. Some silkmoths with larger and slower wings experience large body pitch and vertical oscillations at their flapping frequencies. Such oscillations also cause the wings to rotate with the body in addition to the active rotation about the thorax. This affects the wing angle of attack, which is crucial to flight performance. In this study on silkmoths, we explore whether the body oscillations arise from aerodynamic or inertial forces, and how much do they affect the aerodynamic force and power in steady forward flight. Across our selection of four wild silkmoth species, we digitized the wing morphology and steady forward flight kinematics from a total 15 wingstrokes of 9 individuals. To systematically evaluate the aerodynamic consequences of wing flapping to body oscillation coupling, we use a quasi-steady blade element model with fully time-varying body and wing kinematics. We find that body oscillations are a consequence of both aerodynamic and inertial forces from the wings. In addition, they directly affect the wing angle of attack within a wingstroke and thus, significantly impact the wingstroke-averaged aerodynamic forces, reduce profile power, and improve body pitch stability. These results might have strong implications for the design of flapping-wing micro air vehicles (FW-MAVs) that require high stability and low power requirements while maintaining agility.

Abstract N° 54

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The biophysics of the bacterial hypoosmotic response

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Bacteria have little control over their environment and, for this reason, they possess powerful adaptive mechanisms such as osmoadaptation. In high osmolarity environments, bacteria accumulate intracellular osmolytes, but under a sudden osmotic down-shock, they release small metabolites, constituting ~10 % of their dry weight, within a mere 20 ms. The release system comprises two major mechanosensitive channels, MscS and MscL, that are directly activated by tension in the cytoplasmic membrane. The larger MscL activates at near-lytic tensions, whereas the smaller MscS opens at moderate tensions and closes last to terminate the immense permeability response. Here we overview key aspects of the osmolyte release response: the kinetics of bacterial cell swelling and osmolyte release, the structures of MscS and MscL, the physical parameters of gating, and models of the gating processes. We will focus on several important features of these channels, including the special pore hydration properties that allow these channels to function as leakproof valves under resting conditions in cells energized to -200 mV, the positions of activating barriers that define the transition rates, and selectivity mechanisms. The last part will emphasize the critical capacity of MscS to proceed from the closed to the inactivated state, which is non-conductive and tension-insensitive. Inactivation occurs under moderate tension near the MscS activation threshold and is sharply augmented by both increased crowding pressure in the cytoplasm and by depolarizing voltages. Dissection of the dynamics of the entire MscS-MscL channel population using patch-clamp shows that osmolyte release likely proceeds in two phases. After membrane tension peaks, the entire open channel population reduces tension down to the MscL threshold resulting in MscL closure still accompanied by residual flickering activity due to the non-inactivating behavior of MscL. The MscS population, which is fully open at that pressure, subsequently drives membrane tension far below the MscL threshold thus silencing the large channel. When MscS reaches its own threshold, it enters the silent inactivated state, thus terminating the permeability response. Because MscS is prompted to inactivate by near-threshold tension, membrane depolarization, and increased concentration of internal impermeable osmolytes, it does act as a perfect safety valve with three separate inputs enabling the self-shutting-off capacity.

Abstract N° 55

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Quality assessment of computational models of protein-protein interfaces

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Protein-protein interactions are essential in many biological processes ranging from cellular signalling to enzymatic activity. Incorrect interactions such as protein aggregation can result in neurological diseases and other disorders. In this work, we describe novel methods to determine whether computational models of protein-protein interfaces resemble experimentally determined protein-protein interfaces. We first constructed a dataset of high-quality crystal structures of heterodimers and generated computational models of each complex using state-of-the-art docking methods, including ZDOCK, HDOCK, and RosettaDock. We focus on several features of the core regions of the interfaces to classify the computational models, e.g. the number, atomic overlap energy, hydrophobicity, and packing fraction of the core residues at the interfaces. Using these features, we find important differences between the interfaces of experimentally determined structures and those in poor-quality computational models, or decoys. Based on these results, we developed a machine-learning classifier that can predict the quality of computational models even for interfaces that have not been studied experimentally.

Abstract N° 56

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Unraveling interactions in biomolecular condensates by atomistic molecular dynamics simulations

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Membraneless organelles are dynamical cellular condensates formed via biomolecular liquid–liquid phase separation (LLPS) of proteins and RNA molecules that are currently recognized to play an important role in cellular organization and functioning. Their assembly depends on the formation of a network of transient, multivalent interactions between flexible scaffold proteins and nucleic acids. Despite the blooming research activity in this field, the structural characterization of these entities is limited, and how the phase behavior is encoded in the sequences of the scaffolding biomolecules is still not clear. Understanding how protein and RNA sequences determine these interactions and ultimately regulate the phase separation is an open key challenge. In vitro studies have revealed how specific mutations of aromatic and charged amino acids, which are enriched in biomolecular condensates, alter the propensities to drive the LLPS of protein/protein and protein/RNA mixtures. We used all-atom molecular dynamics (MD) simulations to shed light on the amino acid grammar of biomolecular phase separation taking advantage of a divide-and-conquer approach which allowed us to probe intermolecular interactions in conditions mimicking real condensates. We focused here on two prototypical systems of protein/protein and protein/RNA condensation, namely the N-terminal disordered region of the DEAD-box

helicase 4 (NDDX4) [1] and mixtures of polyuracyl oligonucleotides with either polylysine or polyarginine peptides [2], in order to provide a high-resolution picture of the molecular mechanisms underlying phase separation. Our results show how the structural properties of aromatic amino acids and arginine enable the formation of multivalent interactions, which stabilize the assembly of protein/protein and protein/RNA condensates.

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Abstract N° 57

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Contact map dependence of a T cell receptor binding repertoire

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The T-cell arm of the adaptive immune system provides the host protection against unknown pathogens by discriminating between host and foreign material. This discriminatory capability is achieved by the creation of a repertoire of cells each carrying a T cell receptor (TCR) specific to non-self antigens displayed as peptides bound to the major histocompatibility complex (pMHC). The understanding of the dynamics of the adaptive immune system at a repertoire level is complex, due to both the nuanced interaction of a TCR-pMHC pair and to the number of different possible TCR-pMHC pairings, making computationally exact solutions currently unfeasible. To gain some insight into this problem, we study an affinity-based model for TCR-pMHC binding in which a crystal structure is used to generate a distance-based contact map that weights the pairwise amino acid interactions. We find that the TCR-pMHC binding energy distribution strongly depends both on the number of contacts and the repeat structure allowed by the topology of the contact map of choice; this in turn influences T cell recognition probability during negative selection, with higher variances leading to higher survival probabilities. In addition, we quantify the degree to which neoantigens with mutations in sites with higher contacts are recognized at a higher rate.

Abstract N° 58

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Understanding Drug Resistance through Molecular Simulation

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Understanding how mutations can render a drug ineffective is an important problem for the development of pharmaceuticals, especially those that treat cancer. Drug resistance can be conferred through mutations that

lower binding affinity but luckily these cases can be caught using standard methods such as free energy perturbation. A far more elusive case is mutations that decrease drug residence time with negligible effects on binding affinity. We investigate one such mutation in Abl Kinase, a protein implicated in chronic myeloid leukemia, bound to the drug Gleevec where the standard residence time is 26 minutes, a timescale that would take millions of years to simulate with classical molecular dynamics. We investigate this system using specialized simulation methods that have yielded accurate residence times across 10 orders of magnitude. Our simulations of Abl Kinase are able to recreate experimental results yielding a threefold change in residence time without a significant change in binding affinity. Furthermore, we use our simulations as a computational microscope to explain the mechanism of drug resistance: an unbinding pathway differentiation between the mutant and wild type.

Abstract N° 59

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Sterically confined rearrangements of SARS-CoV-2 Spike protein control cell invasion

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Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is highly contagious, and transmission involves a series of processes that may be targeted by vaccines and therapeutics. During transmission, host cell invasion is controlled by a large-scale (200–300 Å) conformational change of the Spike protein. This conformational rearrangement leads to membrane fusion, which creates transmembrane pores through which the viral genome is passed to the host. During Spike-protein-mediated fusion, the fusion peptides must be released from the core of the protein and associate with the host membrane. While infection relies on this transition between the prefusion and postfusion conformations, there has yet to be a biophysical characterization reported for this rearrangement. That is, structures are available for the endpoints, though the intermediate conformational processes have not been described. Interestingly, the Spike protein possesses many post-translational modifications, in the form of branched glycans that flank the surface of the assembly. With the current lack of data on the pre-to-post transition, the precise role of glycans during cell invasion has also remained unclear. To provide an initial mechanistic description of the pre-to-post rearrangement, an all-atom model with simplified energetics was used to perform thousands of simulations in which the protein transitions between the prefusion and postfusion conformations. These simulations indicate that the steric composition of the glycans can induce a pause during the Spike protein conformational change. We additionally show that this glycan-induced delay provides a critical opportunity for the fusion peptides to capture the host cell. In contrast, in the absence of glycans, the viral particle would likely fail to enter the host. This analysis reveals how the glycosylation state can regulate infectivity, while providing a much-needed structural framework for studying the dynamics of this pervasive pathogen.

Abstract N° 60

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Emergent dynamics in high throughput data

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Emory U

Modeling in biology has firmly established itself in cases where it's been possible to reduce a system to a relatively small number of constitutive parts (think of population dynamics models with a few species, or biophysical neural models with a handful of molecular species involved). In contrast, modern experiments often measure activity of thousands of parts of a studied system, such as activities of thousands of neurons, or frequency of thousands of pathogenic genomes. Building computation models at this scale has proven difficult (and maybe not useful), and we lack intuition about how to interpret results of our experiments. Put another way: When should we be surprised by what we see in high throughput recordings? I will show that simple models can explain seemingly surprising results of high throughput experiments in fields as diverse as neuroscience and immunology (my primary focus here will be on neuroscience), and I will argue that success of these models signals emergence of simpler, collective descriptions of complex biological systems. The goal now is to identify those collective degrees of freedom, and how they interact with each other. Overall, this progress raises hopes for building predictive models of the nervous system without neurons, tissues without cells, and evolution without individuals or genes.

Abstract N° 61

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Using Machine Learning for Inference of C. elegans Neural Network Structure from Calcium Fluorescence Time Series

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Inferring neuronal connections from recordings of a large number of individual neurons is an important problem in neuroscience. Calcium imaging is emerging as a useful technique for simultaneous recordings from a large number of neurons of live organisms; however, the data obtained suffers from poor time resolution, making the task of network inference particularly challenging. Here we use a machine learning architecture [1-2] called `reservoir computing' to perform network inference from low temporal resolution, noisy time-series data of network nodal states, and assess the statistical significance of the results. We test the performance of the link inference technique on publicly available calcium imaging time series datasets from C. elegans and use a particular method of generating surrogate time-series that can assess the statistical significance of network links predicted from our technique. We also use synthetic time-series data from coupled Lorenz oscillators to systematically investigate the effects of sampling rate, sampled subnetwork size, and observational noise strength on link inference technique performance and the validity of the statistical tests with the surrogate data that we use. Our results indicate the importance of machine-

learning-based techniques on link inference tasks from time-series data of neural systems.

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Abstract N° 62

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Neural basis for odor-evoked behavior in C. elegans

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Many small animals use their olfactory sense as the primary source of information to navigate their environment. We study the neural basis of olfaction in Caenorhabditis elegans, whose small 300-neuron nervous system allows it to detect a wide range of odors and change its behavior to move towards more favorable conditions. These animals can precisely sense their chemical environment and use probabilistic chemotaxis strategies that adjust their turning and reversal rates based on the gradient of stimulus concentration. We use a pan-neuronally labeled strain to record whole-brain activity in response to pulses of diacetyl as a paradigm to study the circuit-level implementation of probabilistic navigation strategies. Diacetyl is primarily sensed by the chemosensory neuron AWA and is attractive at low concentrations. At high concentrations, it is also sensed by the nociceptive neuron ASH, and it elicits an aversive response. We use odr-10 mutants, defective in AWA response, combined with a variety of concentrations of diacetyl to shift the balance of the response of these two neurons and thus explore a range of attraction to aversion behavioral responses. We use a microfluidics chip to precisely control the concentration and odor delivery conditions while recording pan-neuronal activity in immobilized worms. We find that sensory neurons encode the stimulus identity, and we can predict properties of the stimulus using their activity alone. In contrast, the activity of motor circuit neurons in response to a given stimulus is variable, encoding a bias towards forward motion or reversals and turns. We explore the characteristics of this response and look for circuit-level implementations of probabilistic motor responses to deterministic stimuli, which involve the action of several interneurons. We generate functional maps of brain activity during and in the absence of stimulus. Our study makes progress towards understanding how behavioral commands are generated depending on sensory input and brain states.

Abstract N° 63

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Statistical properties of large data sets with linear latent features

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Biological interaction networks such as neural networks or amino acid sequences in proteins are critical to the functioning of any biological system. Understanding the underlying working principles of these complex networks is a key challenge. One hope is that the advent of high-throughput experiments provides the data necessary to develop this understanding. The trend of modern high-throughput experiments is to simultaneously record data from a rapidly growing number of network units. It is a common belief that many such datasets contain a hidden low-dimensional structure, where the number of true degrees of freedom is (significantly) smaller than the number of recorded units. However, a ground truth for what data observables, such as the distribution of correlations and eigenvalues of the correlation matrix, etc. look like for low-dimensional data is still missing. Here, we build a random matrix model with linear latent features and additive noise which captures essential aspects of low-dimensional data. We compute the statistics of key observables analytically and numerically. Our analysis provides quantitative insight into how the number of latent features affects the shape of observable distributions, how to draw a boundary between signal and noise in the absence of a clear spectral gap, as well as how to judge the sampling quality of data as a function of the model parameters.

Reference:

P. Fleig and Ilya Nemenman, “Statistical properties of large data sets with linear latent features,” arXiv:2111.04641 [cond-mat]

Abstract N° 64

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Recurrent activity in neuronal avalanches

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A new statistical analysis of large neuronal avalanches observed in mouse and rat brain tissues reveals a substantial degree of recurrent activity and cyclic patterns of activation not seen in smaller avalanches. To explain these observations, we adapted a model of structural weakening in materials. In this model, dynamical reduction (or ‘weakening’) of neuron firing thresholds closely replicates experimental avalanche size distributions, firing number distributions, and patterns of cyclic activity. This agreement between model and data suggests that a mechanism like dynamical threshold reduction plays a key role in recurrent activity found in large neuronal avalanches and we thus expect these results to illuminate the causes and dynamics of such large avalanches.

Abstract N° 65

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Monitoring DNA organization and transcription in single cells using multiplexing microscopy

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DNA is organized at multiple length scales, from nucleosomes to chromosome territories. However, it is at intermediate levels of organizations that tissue-specific transcriptional regulation takes place. We will investigate this regulation during the early stages of differentiation in Drosophila embryonic development. For this, we have developed new imaging methods that rely on the use of micro-fluidics to perform sequential and combinatorial acquisition of tens of different species in single cells. These techniques revealed the existence of activating or repressing enhancer hubs that provide a scaffolding for the regulation of transcription during early embryogenesis.

Abstract N° 66

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Identification of the source of extrinsic noise from the stochastic dynamics of gene expression

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Cell-to-cell variability in protein concentration for highly expressed genes is mostly dominated by extrinsic noise. Extrinsic noise can be due to fluctuations of several possible cellular factors, such as the cell volume or the level of key enzymes in the expression process. However, how to identify the specific dominant sources of extrinsic noise in empirical systems is still an open area of research.

This work considers a general stochastic model of gene expression with extrinsic noise represented as colored fluctuations of the model parameters, and focuses on out-of-equilibrium system properties.

More specifically, we consider both the process of activation and of inactivation of a gene of interest, and show that qualitatively different protein variability trends can be identified depending on the specific fluctuating parameters. This result provides the theoretical basis for an experimentally accessible way to pinpoint the dominant sources of extrinsic noise using multiple time-coarse measurements. Moreover, we fully explore how the expression dynamics depends on the intensity and the timescale of extrinsic fluctuations for different sources of extrinsic noise.

Abstract N° 67

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Disentangling phylogeny and optimization contributions in protein sequences

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Determining protein structure and function is an essential task in biology as proteins are at the heart of most key processes in the cell. The structures of proteins are tightly related to their functions, and all the information on the folding of the protein is encoded in the sequence. Moreover, the growing number of protein sequences provides a large dataset, allowing us to apply statistical methods to analyze these sequences, thus shedding light on their structure, function and evolution. In particular, such methods are able to predict protein structure from sequences, and shed light on functional collective modes of correlation in sequence alignments. Among these methods, pairwise maximum entropy models, known as direct-coupling analysis (DCA) have allowed protein structure prediction from sequence data alone, as well as prediction of protein-protein interaction partners. However, some predictions of structural contacts obtained by this method are incorrect. They have been attributed to different causes (phylogeny, structural variations, indirect correlations...) with no real consensus on their origin. Our first aim is to disentangle and understand the possible causes by inspecting synthetic, thus controlled, data. We find notable differences between sequences constrained only by optimization and sequences that also involve phylogeny, i.e. evolutionary history. We characterize the false positive DCA contacts originating from these two sets of data. We find that they present different properties, such as network characteristics or conservation. Importantly, we find that specific false positives are associated to phylogeny and do not reveal any information about contacts or network structures. Next, we studied the contributions of phylogeny and natural selection to the inference of interaction partners from protein sequences. We showed that phylogenetic correlations and the associated false positive couplings are key to partner prediction performance in synthetic and real data. Indeed, partner prediction performance is increased in strongly phylogenetic data due to these phylogenetic couplings. This stands in contrast with the opposite behaviour observed in contact prediction.

Abstract N° 68

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Weak tension accelerates hybridization and dehybridization of short oligonucleotides

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The hybridization and dehybridization of DNA subject to tension is relevant to fundamental genetic processes and to the design of DNA-based mechanobiology assays. While strong tension accelerates DNA melting and decelerates DNA annealing, the effects of tension weaker than 5 pN are less clear. In this study, we developed a DNA bow assay, which uses the bending rigidity of double-stranded DNA (dsDNA) to exert weak tension on a single-stranded DNA (ssDNA) target in the range of 2-6 pN. Combining this assay with single-molecule FRET, we measured the hybridization and dehybridization kinetics between a 15 nt ssDNA under tension and a 8-9 nt oligo, and found that both the hybridization and dehybridization rates monotonically increase with tension for various nucleotide sequences tested. These findings suggest that the nucleated duplex in its transition state is more extended than the pure dsDNA or ssDNA counterpart. Our simulations using the coarse-grained oxDNA2 model indicate that the increased extension of the transition state is due to exclusion interactions between unpaired ssDNA regions in close proximity to one another. This study highlights an example where the ideal worm-like chain models fail to explain the kinetic behavior of DNA in the low force regime.

Abstract N° 69

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Traffic on the genome: Group dynamics of RNA polymerases during gene transcription via DNA supercoiling

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The flow of genetic information from DNA to messenger RNA and to protein is one of the fundamental processes in biology. RNA polymerase is a molecular motor that is in charge of the first step of this process, i.e. transcription of DNA into mRNAs. Inside cells, multiple copies of RNA polymerases translocate along the same DNA template concurrently, and whether they exhibit any group dynamics, similar to other well-known molecular motors, remains unclear. Using single-molecule mRNA counting, we measured the relationship between the density of RNA polymerases and their speed, and our results suggest long-distance cooperation between the polymerases. Interestingly, we found that the cooperation can turn into antagonism when the flux is stopped at the promoter. Our model suggests that these two contrasting modes of transcription are mediated by the mechanical stress associated with DNA supercoiling. We found the transmission of the mechanical stress can be effective more than 2 kb within and between genes, suggesting that it is an important mechanism to control the RNA polymerase traffic on the genome.

Abstract N° 70

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Unravelling individual transcription events and transcription bursting mechanisms by signal deconvolution in RNA tagging systems

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RNA tagging is a powerful technique for studying transcription in live cells. We present a novel machine learning method that allows for the first time to precisely detect the transcription initiation events in single nuclei. Using this information, we determine, for a population of nuclei, the number of hidden epigenetic states involved in transcriptional bursting. The method also allows to identify and compare multiple bursting

mechanisms. To support its wide utilisation, we offer two implementations of the method as Python and Matlab tools, that include a graphical interface. The comparison of our tools to alternative methods emphasizes their advantages in terms of precision, execution time and flexibility. Our method is rapid and accurate. By its unique features such as the determination of the number of states and the simultaneous analysis of several

potential gene bursting models, signal deconvolution is an ideal analytic framework for RNA tagging experiments. We have applied this method to very different biological problems: transcription during blastoderm stages of Drosophila's development, latency of HIV promoters in human cells, etc.

Abstract N° 71

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Statistical Analysis of Sharing in Antibody Repertoires

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Adaptive immunity success relies on the extraordinary diversity of protein receptors on B cells membranes. Recent progress in deep sequencing methods has been followed by the development of probabilistic models characterizing repertoire sequence distributions [1]. Here we present a statistical approach defined in terms of a probabilistic V(D)J recombination model enhanced by a selection factor that describes repertoire diversity and that predicts with high accuracy its level of publicness, i.e. the number of sequences that will be shared between any number of individuals. While the model shows perfect agreement with healthy repertoires it clearly underestimates sharing between repertoires affected by a common antigen. This deviation is a sign of a stronger antigen driven selection that opens the possibility of finding antigen-specific antibodies.

Abstract N° 72

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Landscapes of Genomic Architecture Across Evolution

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The human genome is composed of 46 DNA molecules — the chromosomes — with a combined length of about two meters. Chromosomes are stored in the cell nucleus in a very organized fashion that is specific to the cell type and phase of life; this three-dimensional architecture is a key element of transcriptional regulation and its disruption often leads to disease. What is the physical mechanism leading to genome architecture? If the DNA contained in every human cell is identical, where is the blueprint of such architecture stored?

In this talk, I will demonstrate how the architecture of interphase chromosomes is encoded in the one-dimensional sequence of epigenetic markings much as three-dimensional protein structures are determined by their one-dimensional sequence of amino acids. In contrast to the situation for proteins, however, the sequence code provided by the epigenetic marks that decorate the chromatin fiber is not fixed but is dynamically rewritten during cell differentiation, modulating both the three-dimensional structure and gene expression in different cell types.

This idea led to the development of a physical theory for the folding of genomes, which enables predicting the spatial conformation of chromosomes with unprecedented accuracy and specificity. Finally, I will demonstrate how the different energy terms present in the model impact the topology of chromosomes across evolution. Our results open the way for studying functional aspects of genome architecture along the three of life.

Abstract N° 73

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Optimal checkpoint strategies balancing risk and speed

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Checkpoints arrest biological processes allowing time for error correction. The phenomenon of checkpoint override during cellular self-replication is biologically critical but currently lacks a quantitative, functional, or system-level understanding. To uncover fundamental laws governing error-correction systems, we derived a general theory of optimal checkpoint strategies, balancing the trade-off between risk and self-replication speed. Mathematically, the problem maps onto the optimization of an absorbing boundary for a random walk. We applied the theory to the DNA damage checkpoint (DDC) in budding yeast, an intensively researched model checkpoint. Using novel reporters for double-strand DNA breaks (DSBs), we first quantified the probability distribution of DSB repair in time including rare events and, secondly, the survival probability after override. With these inputs, the optimal theory predicted remarkably accurately override times as a function of DSB numbers, which we measured precisely for the first time. Thus, a first-principles calculation revealed undiscovered patterns underlying highly noisy override processes. Our multi-DSB measurements revise well-known past results and show that override is more general than previously thought.

Abstract N° 74

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Impact of mechanical constrains on cells rheology and physiology

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Cells are complex systems composed of a great number of components with different sizes and properties (organelles, proteins, ribosomes, etc.) therefore the interior of a cell is a crowded environment, even though many reactions take place there. Our study focuses on what happens when a cell is constrained by a mechanical stress, with a particular interest on the crosstalk between cell crowding and physiology.

To impose a mechanical stress we used PDMS chips in order to confine cells inside chambers with an import of culture medium. Our chambers can be closed by cells themselves or by applying a pressure on a membrane. (Delarue et al., 2016) When cells grow inside this confined chamber, they apply a mechanical pressure on their neighbor, and a growth-induced-pressure emerges. We can assess the pressure by measuring the deformation of PDMS. We also used osmotic stress by adding sorbitol in culture medium.

We analyzed the diffusion of nanoparticles inside cells to study crowding. To do this we genetically-modified the yeast S. cerevisiae to induce fluorescent nanoparticles called GEMs (Delarue et al., 2018). We observed that diffusion decreased when mechanical stress increased. In parallel, for the physiological part, we chose to look at the production rate of fluorescently labeled proteins. Similar to the diffusion, we saw that the production rate decreased with the increase of pressure or osmotic stresses.

We showed that reaction rates inside cells slowed down with increasing crowding when cells are under mechanical pressure or osmotic stress and so we imagine

a physical feedback model who can predict that crowding limit growth under pressure (Alric et al., 2022).

In general, if reactions are limited by crowding which is a generic biophysical parameter, we can easily imagine that other physiological aspects (metabolism, growth) can be affected by a similar mechanism, which could occur in various organisms (e.g. bacteria, mammalian cells).

Abstract N° 75

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Coordination of chromosome replication/segregation and cell-cycle progression in bacteria: the role of rapid chromosome movements?

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Cell-cycle control is vital for cells across all kingdoms of life as it ensures that key processes of cell physiology are brought to completion in the right way. In bacteria, despite over 60 years of active research and outstanding progress[1], we are still far from a complete and coherent picture of how this control takes place.

One key open question is how chromosome replication/segregation and cell-cycle events (prominently cell division) are correlated. It has been suggested that cell-cycle progression may be coordinated mechanically by conformational changes of a "Progression Control Complex"[2]. However, to what extent these rapid chromosome movements play a role in cell-cycle progression remains unclear.

The main difficulty in fully addressing these kinds of questions is the need of high-throughput dynamic single-cell data, which only recently have become routinely available. These measurements can potentially access the full correlation pattern of several cell-cycle events, thus making mechanistic interpretations possible, and indeed some work have been put along this direction by different groups in the past years.

I will review the major highlights of the "parallel decisions" paradigm, a recent logic put forward by my group, according to which cell-cycle may progress through the integration of parallel decisions. For example, it has been shown that, in E.coli, both replication-dependent and replication-independent processes concur to set cell division on competing time-scales[3,4].

I will also show preliminary results regarding (A) the possible integration of "parallel decisions" at the onset of constriction, as recently suggested[5], and (B) quantitatively testing the "Progression Control Complex" hypothesis.

My work includes mathematical models and new single-cell experiments that I performed in the Cicuta Lab at the Cavendish Laboratory, where cell-cycle progression and rapid loci movements are tracked for the first time together over multiple cell-generations.

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Abstract N° 76

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Characterization of colon tissues stiffness during carcinogenesis by atomic force microscopy

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In addition to standard clinicobiological parameters used for colorectal cancer (CRC) patient management, new features are needed in clinical practice to improve both progression-free survival and overall survival of patients. In some cancers, the mechanical response due to the microenvironment can contribute to cancer progression, metastasis or constitute physical barriers for treatments or immune infiltration. However, these physical properties are still poorly known for colon tissues. Using a pluridisciplinary approach including clinics, physics and geostatistics, we characterized the stiffness of CRC during carcinogenesis. For this, we performed µm scale analysis by Atomic Force Microscopy (AFM) on a prospective cohort of 18 colon adenocarcinoma. For some patient, both a chirurgical sample of normal “at distance” tissue and of tumor tissue, naïve of treatment were selected. Optimizations were performed to preserve tissues mechanical properties and to correlate both AFM and optical microscopy regions of interest. We compared the normal epithelium at the distance with the proximal sometimes present in tumors areas. We also compared the stiffness differences between malignant epithelial cells and stroma rich regions in some tumors. Our data show that normal epithelium is softer than tumors. In tumors, the stroma is stiffer than the malignant epithelial cell regions. Among the clinicobiological parameters, left location and RAS mutations were found to be associated with increased stiffness. Thus, measuring tumor rigidity of CRC tissues may open new avenues for CRC treatment.

Abstract N° 77

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Localized growth drives pore space patterning in mesophyll development

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Morphogenesis, or the emergence of spatial structure and heterogeneity during biological development, is driven by changes to cell behavior which generate both patterns and function in young tissues. A key player in the morphogenesis of plants is schizogeny, or the formation of pore space by breaking of adhesive cell contacts. Here, we study the poorly understood case of schizogeny during the development of spongy mesophyll, or the interstitial tissue of leaves and flowers. We develop a two-dimensional computational model using deformable

polygons that recreates the developmental process from confluent mesophyll progenitor cells to porous, network-like mature cells with highly non-spherical shapes. We show that localized growth of cell perimeter drives schizogeny, shape change, and ultimately the increase in tissue porosity necessary for development. While the specific microstructure of mature networks can be tuned by variations to cell mechanical parameters, we find that the network formation process is robust and can occur in a variety of conditions. Comparison with experimental observations of spongy mesophyll in leaves from Arabidopsis thaliana, both early in development and in mature samples, reveals that our model recapitulates the developmental and structural features of this complex tissue.

Abstract N° 78

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A mechanistic theory of bacterial resource allocation out of steady state.

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The currently available theories that describe bacterial growth physiology across environmental changes are are typically phenomenological. For example, the standard out-of-equilibrum framework is provided by the "Flux Controlled Regulation" (FCR) model (Erickson et al. Nature 2017), which makes use of a steady-state assumption linking ribosomal activity and regulatory strategy, and does not describe ppGpp-mediated nutrient sensing and transcriptional regulation of ribosomal operons. Classic attempts to formulate mechanistic theories have not been integrated with the growth laws framework. To highlight the need for such theories, the time scales involved in the shifts are complex and both recent and early reports show the emergence of multiple time scales and damped oscillatory behavior. In this work, we present a new model which integrates the classic FCR framework with the explicit description of (i) the amino-acid pool (ii) ppGpp sensing of uncharged tRNAs and (iii) trascriptional programming of the protein allocation strategy. This theory reproduces the latest data on steady state growth as well presenting the experimentally observed complexity of relaxation time scales. The gained mechanistic insight unlocks possibility to predict of the results of specific perturbations related to the ppGpp circuit.

Abstract N° 79

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Hole closure in sheets of active deformable particles

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Holes created in sheets of epithelial cells provide an in vitro experimental system to gain insight into wound healing in biological tissues. Experimental studies have found that holes close as a result of collective cell motion that is influenced by cell shape, motility, and adhesion. We propose the deformable particle (DP) model to study the mechanics of hole closure in epithelial sheets, where the cells are modeled as individual deformable polygons

with Nv vertices whose positions are determined by a shape-energy function with terms that control the cell area, perimeter, and curvature. To model the dynamics of hole closure, we include adhesive protrusions and collective contractile tensions for cells along the perimeter of the hole. An important advantage of the DP model for describing hole closure is the ability for individual cells to take on a wide range of complex shapes. We vary the cell adhesion, shape parameter, rate of protrusion, and contractile tension to determine how they affect collective cell motion and the rate of hole closure.

Abstract N° 80

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Extrinsic and intrinsic transcriptional noise in the control of HIV-1 latency

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HIV latent reservoirs are cells infected with HIV-1 that remain in the body for a long period. These cells express minimally and remain undetected by the host immune system. Stochastically expressing latent cells are hypothesized to be the key drivers of the infection and viral rebounds in HIV patients. From previous studies we understand that this viral rebound is because of the latently infected cells being reactivated. However, the mechanism is not well understood. Latent cells are not completely inactive. They produce small bursts of activity that cannot be detected but can potentially lead to viral rebound. Every now and then a burst of viral particles activates the feedback loop that drives efficient HIV-1 transcription and triggers latency exit. This stochasticity is believed to be arising at the level of promoter switching dynamics as a result of two types of transcriptional noise; extrinsic fluctuations (regulated by environmental factors affecting all the gene) and intrinsic fluctuations (the inherent randomness of the reaction system). Studying transcriptional noise can help us understand latency better.

We aim to study the stochastic fluctuation of HIV-1 transcription and separate the contribution of extrinsic and intrinsic factors. In this study an HIV-1 reporter labeled with the MS2 tag will be integrated in multiple independent copies into a reporter cell line (initially as a homozygous CRISPR integrated in HCT116 diploid cells, then as multiple random viral integrations in Jurkat T cells). We will perform live cell imaging using microfluidic devices that will allow us to image 100-1000 cells and generate high-throughput data. The reporter copies sharing the same cellular environment would allow us to follow correlated (extrinsic) as well as uncorrelated (intrinsic to the gene network) fluctuation. The high-throughput live cell imaging data describing the viral transcription will be analyzed using image analysis softwares. We will use a novel autocovariance method to distinguish between intrinsic and extrinsic noise and extract the respective timescales. The statistics of the transcription events will also be extracted using a pipeline developed in the lab. These will then be used to determine the role of transcriptional noise in triggering viral bursting by mathematical modelling of the promoter dynamics. Our study will help determine the respective importance of extrinsic and intrinsic noise for exit from viral latency.

Abstract N° 81

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Competition between actin treadmilling and myosin contractility regulates the formation of actomyosin rings

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Actin filaments in eukaryotic cells form a shell-like actin cortex under the plasma membrane and control cell shape, exert mechanical forces and participate in signaling pathways. The actin cortex is highly polymorphic, adopting diverse forms such as the ring-like structures found in podosomes, axonal rings, and immune synapses. While actin rings and cortices have been widely observed and studied, the biophysical principles underlying the formation of these structures remain unknown. We have used a combination of experimental measurements on live T cells and computational studies performed using a molecular simulation platform called MEDYAN to address this question. We studied the effect of actin filament treadmilling rates and myosin activity on the stability of the actin ring at the T cell immune synapse. Experimentally, we showed that inhibiting myosin activity in Jurkat T cells caused the actin ring to become disordered, leading to a less well-defined ring-like structure. Disrupting filament treadmilling caused the disintegration of actin rings into clusters, while enhancing myosin activity induced centripetal collapse of pre-existing rings. These observations are consistent with simulations which demonstrated that actomyosin networks condense into clusters at low treadmilling rates or high myosin concentrations but form ring-like structures at high treadmilling rates and low myosin concentrations. Our findings suggest that the ring-like actin structure formed at high treadmilling rates is the preferred overall low energy state, and that myosin activity regulates the stability of the actin ring.

Abstract N° 82

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Developing a coarse-grained model for secretion of the passenger domain of the autotransporter EspP

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Gram-negative bacteria are enveloped by two membranes with a periplasm in between. There is no typical source of energy, such as ATP, between the two membranes, so these bacteria have evolved several self-sufficient pathways for secreting proteins across the outer membrane. The Type V(a) secretion system, also known as autotransporters, is one of the simplest among them, consisting of a translocator domain in the outer membrane and a passenger domain that is secreted to the extracellular milieu. EspP is one example of an autotransporter that secretes virulence factors with the help of the β-barrel assembly machinery (BAM) complex. However, the mechanism of secretion is still poorly understood. To elucidate this mechanism, we built a coarse-grained model for the passenger domain during secretion through a hybrid BAM-EspP system. We also use electrostatic potential maps generated from atomistic simulations to include realistic electrostatic interactions from the proteins and the outer membrane. We observed that the passenger domain has a higher possibility to secrete in the presence of a negatively charged membrane. Additionally, different regions of the passenger domain show different sensitivities to the electrostatic environment.

Abstract N° 83

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Towards Cell Scale Simulations of the Mechanics and Chemistry of the Cytoskeleton and Membrane.

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The cytoskeleton is an active polymer network that consumes chemical energy to organize and do mechanical work. In connection with complex chemical networks in the cytoplasm and cell membrane; the cytoskeleton can create many large scale dynamic structures including, filopodia, lamellipodia, actin waves, and actin rings. We will discuss how to simulate the chemical and mechanical interactions of actin filaments, crosslinkers, myosin motor proteins, and membranes at the micrometer length scale and minute time scale using the coarse grained stochastic MEDYAN model. We will also discuss how the new version of MEDYAN written in the Julia programming language will enable whole cell scale simulations and more intricate molecular motor models. Currently, a prototype written in Julia is 10x faster than the current C++ version of MEDYAN in simple reaction diffusion simulations.

Abstract N° 84

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Searching for the Mechanism of Pattern Formation in Proteus mirabilis

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Proteus mirabilis is notable among bacteria for its superior swarming ability, but it is equally notable for its counterintuitive swarming pattern. Rather than expanding continuously, P. mirabilis alternates between phases of swarming and consolidation, generating colonies with a striking concentric ring pattern. Why P. mirabilis stops its swarming and enters a consolidation period is unknown, as is the mechanism by which it coordinates a stop to swarming. Other bullseye-pattern formation systems, such as chemotaxis or cellular clocks, have not been able to explain the P. mirabilis swarming cycle, so we are examining the collective motion of swarmer cells to elucidate the role of physical mechanisms in their pattern formation. P. mirabilis is a unique system of active matter, as its swarmer cells have a high aspect ratio and flexibility and engage in collective motion at high densities. We're using advanced microscopy to characterize the dynamics of this active matter system and to bridge the gap between single cell dynamics and collective swarming behavior in order to understand how and why swarming ends and consolidation begins.

Abstract N° 85

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Structural and dynamics properties of protein/RNA condensates by multiscale molecular simulations and small-angle scattering experiments

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Membraneless organelles (MLOs) are liquid-like condensates, typically rich in proteins and RNA, which are formed via liquid-liquid phase separation (LLPS). In particular, DEAD-box proteins (DDXs) constitute a large family of RNA-helicase proteins involved in a number of different processes in RNA metabolism and they have been shown to regulate RNA-containing condensates in vitro and in vivo. DDXs are composed of a two-domain folded core that can bind to RNA in an ATP-dependent fashion manner and of two disordered termini which can form dynamic protein-protein interactions, necessary for condensate formation.

This complex network of intermolecular interactions scaffold MLOs and it is finely controlled by ATP hydrolysis. To investigate these condensates as non-equilibrium systems and to characterize the consequences of energy consuming processes at the molecular level, Small-angle X-ray and/or neutron scattering (SAXS/SANS) experiment and multiscale molecular dynamics (MD) simulations will be combined. In particular, scattering experiments in dilute phase and accurate atomistic MD simulations will be exploited to draw an effective multi-state model for DDXs which describes the most important states occurring in RNA binding and ATP hydrolysis. This coarse-grained model will be used together with complementary SAXS/SANS measurements on condensates to efficiently investigate the structural, dynamical and functional properties of these large-sized assemblies.

Abstract N° 86

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The onset of Thrombosis in Deep Veins (DVT): The coupling effect of stretch and shear stresses on endothelial cells

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Deep vein thrombosis is frequent and can be life threatening when the thrombus is breaking. It is thus important to better understand its formation and to control thrombus stability. It generally forms in venous valves. We studied the onset of Thrombosis and investigated the dynamics of venous valves using in vitro models that mimic the pathology. We start by exposing Human endothelial vein cells (HUVEC) to mechanical constraints as found in the valve and to test the relationship between the mechanical constrain and the pro/anti-thrombotic properties of the endothelial monolayer by labelling of proteins of interest like thrombomodulin. For this, we first set up a protocol to stretch under cyclic uniaxial extension HUVECs 2D monolayers. In a second time, we developed a microfluidic device to mimic the hemodynamic conditions of the valvular sac and to explore the combined effect of both flow and stretching stresses. Taken together, these two approaches will facilitate quantitative study of mechanical strain and endothelial cells interactions, developing into an organ-on-chip device to better understand deep vein thrombotic diseases.

Abstract N° 87

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Evaluating Nanoparticle Delivery during Normothermic Machine Perfusion of Human Liver

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Introduction: Every day, deceased donor livers are recovered but then declined for transplant while people die or become too sick for transplant. Many of these organs go unused due to the limitations in current cold storage techniques. Ex vivo normothermic machine perfusion (NMP) has emerged as an alternative approach that provides the potential for pre-transplant repair of injured organs, but the optimal therapeutic interventions remain unclear. Previous research from our lab has suggested that polymeric nanoparticles (NPs) provide an ideal solution given their adaptability and capacity for prolonged therapeutic effect post-transplant. However, we have also shown in NMP of human kidney that fibrinogen-mediated red blood cell obstructions can impede drug delivery and impair the ability to distribute NPs. Pretreatment with fibrinolytic agents can prevent this NP aggregation. In this work in progress, we sought to assess if a similar phenomenon impacts the delivery of nanomedicines within transplant-declined human liver undergoing NMP.

Methods and Results: 3 human livers declined for transplant were perfused on a clinical perfusion device (OrganOx Mextra) for 12 hours. Poly-lactic acid-Poly-ethylene glycol NPs loaded with DiI dye were administered in the perfusate. Tissue biopsies and serum samples were collected at several time points after NP injection. We observed that liver function relates to serum NP concentration. The poorer functioning livers had an initial rapid decrease in NP concentration in the first hour post-NP injection, while all livers showed a slow decrease over the next several hours of perfusion. The poorest functioning liver was the only liver to have 50% clearance of NPs from the serum, about 8 hours post-NP injection. In each liver, evidence of red blood cell aggregates was observed.

Conclusions: Our preliminary analysis suggests NPs did accumulate at sites of obstruction, however there appear to be substantially fewer red blood cell and NP aggregates in this cohort of human livers as compared to our previous observations in human kidney. Despite relating serum NP concentration and liver function, the liver biopsy data shows a limited amount of NP aggregation. Therefore, we likely do not need to pretreat the livers with fibrinolytic agents as done in the kidney. We can apply this knowledge going forward to try to optimize NP targeting during NMP for maximum therapeutic effect in the liver.

Abstract N° 88

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Characterizing Evolutionary and Physical Energy Landscapes of Evolving, Devolving, and Random Sequences

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Due to evolutionary constraints, protein sequences cooperatively fold into unique, energetically minimized three-dimensional structures in timescales as short as microseconds. The evolutionary importance of maintaining a stable protein structure can be quantified by the selection temperature. The selection temperature can be calculated using AWSEM, a coarse-grained free energy function, and DCA, a protein-specific information theoretic Hamiltonian. Expanding upon the list of previously studied proteins, we quantified selection temperatures of multiple proteins with a wide-array of biological functions. We found selection temperatures to be below physiological temperatures, indicating that folding energetics are an important evolutionary consideration. Furthermore, we incorporated temporal information by calculating the energetics of pseudogenes, former protein-coding genes that have devolved due to random mutations over time. We found that pseudogenes are more unstable than protein sequences, as a function of their degree of devolution and evolutionary time.

Abstract N° 89

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Emergent statistical laws in single-cell transcriptomic data

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Large scale data on single-cell gene expression have the potential to unravel the specific

transcriptional programs of different cell types. The structure of these expression datasets suggests a similarity with several other complex systems that can be

analogously described through the statistics of their basic building blocks.

Transcriptomes of single cells are collections of messenger RNA abundances transcribed

from a common set of genes just as books are different collections of words from a

shared vocabulary, genomes of different species are specific compositions of genes

belonging to evolutionary families, and ecological niches can be described by their

species abundances. Following this analogy, we identify several emergent statistical laws

in single-cell transcriptomic data closely similar to regularities found in linguistics,

ecology or genomics. A simple mathematical framework can be used to analyze the

relations between different laws and the possible mechanisms behind their ubiquity.

Importantly, treatable statistical models can be useful tools in transcriptomics to

disentangle the actual biological variability from general statistical effects present in

most component systems and from the consequences of the sampling process inherent to

the experimental technique.

Abstract N° 90

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Multi-strain phage-induced clearance of Pseudomonas aeruginosa

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The evolution of antibiotic resistant bacteria is a serious global health threat. Phages are increasingly considered as a therapeutic alternative to antibiotics to treat bacterial infections. One of the biggest challenges in phage therapy is that bacteria can develop resistance mechanisms against phages during treatment, leading to adverse clinical outcomes. To overcome this obstacle, it is possible to apply cocktails of multiple phages hoping that bacteria will not be able to develop resistance against all of them. On the other hand the underlying molecular and eco-evolutionary mechanisms governing the emergence of phage resistance during therapy are not fully characterized, and therefore there is no robust pipeline to design phage cocktails.

In our work we address quantitatively how to design a cocktail of phages efficient against infections by the bacterium Pseudomonas aeruginosa.

We select two phages that infect Pseudomonas aeruginosa through two different adsorption routes, hypothesizing that this will reduce the odds of the emergence of resistance against both of them.

We characterize the efficacy of a cocktail with these two phages in clearing bacterial populations, through in vitro experiments and computational models.

In parallel, we study a theoretical model of an infection in an immunocompromised host, where the available immune resources and phages act in synergy against bacteria that can develop phage resistance. We quantify the therapy outcome whether a single phage or two phages are applied, as a function of the phage host range and of the immune system strength.

Abstract N° 91

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Investigating human chromosome organization by whole-genome simulations

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In recent years, both theoreticians and experimentalists have dedicated significant effort to studying chromosome organization and its relation to gene expression and regulation. Many experimental procedures were designed, such as Hi-C or ChIA-PET, to capture structural information of chromosomes. Several theoretical models were proposed to understand the principles underlying chromosome compartmentalization. MiChroM is a model derived using the Maximum Entropy Principle that describes the compartmentalization behavior of interphase chromosomes. Compared to experimental data, the agreement of the in silico contact maps suggested that phase-separation of A/B compartments is a key feature determining the structure of individual chromosomes. In this work, we use the Open-MiChroM software to simulate the whole genome of GM12878 human cell at 50 kb resolution, seeking to understand whether the nuclear organization of all chromosomes would rely on the same principles observed for the organization of a single chromosome. The ensemble of structures generated by our simulations captures chromosome organization within territories and phase separation of chromatin compartments in intra-chromosomal regions, with active chromatin localizing in the periphery of the chromosome territory. Additionally, phase separation of chromatin compartments is observed in the interface between chromosomes, and the periphery of the nucleus is also populated with active chromatin. The intra-chromosomal contact maps are consistent with experimental data, whereas the inter-chromosomal maps present higher contact probabilities for heterochromatin loci. Altogether, these results suggest that, in addition to the chromatin-chromatin interactions, the model needs to consider external features like chromatin interactions with the nuclear lamina to capture the genome organization within the nucleus.

Abstract N° 92

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MscS is an essential component of the bacterial osmolyte release system

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Most free-living microorganisms and environmentally transmitted pathogens are subjected to drastic changes in osmolarity. In high osmolarity environments, bacteria massively accumulate intracellular osmolytes but quickly release them under a sudden osmotic down-shock. The latter mechanism involves MscS and MscL channels that

are directly activated by tension in the cytoplasmic membrane. Initial studies suggested that the complement of mechanosensitive channels was partially redundant, but here we show that MscS fulfills a unique role as a 'smart' terminator of the osmotic permeability response. Using a set of chromosomal deletion mutants we show that: (1) in rapid dilution stopped-flow experiments both MscS and MscL comparably mediated fast osmolyte release from cells, however parallel survival assays showed that MscS by itself was capable of fully rescuing the population, whereas MscL was unable to rescue and even became toxic in the absence of MscS and MscK; (2) in contrast to MscL, MscS has the capacity to inactivate at tensions near its own activation threshold, and the rate of this inactivation is increased in the presence of cytoplasmic crowding agents; (3) MscS inactivation and recovery are voltage-dependent, with deeper inactivation and slower recovery at depolarizing potentials and less inactivation and faster recovery at normal hyperpolarizing voltages; (4) patch-clamp experiments performed with pressure protocols mimicking the tension changes during osmotic shock revealed a functional interplay between the two channels. We observe that the MscL population residing at its own high-tension threshold closes incompletely and leaks. Meanwhile, the low-threshold MscS population stays open and is able to reduce tension far below the MscL threshold such that MscL becomes silent. MscS, prompted to inactivate by near-threshold tension, depolarized membrane potential, and increased concentration of impermeable osmolytes inside the cell acts as a perfect safety valve properly terminating the hypoosmotic permeability response.

Abstract N° 93

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Coarse-grained Molecular Dynamics Simulations of Ca2+-Calmodulin

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Calmodulin (CaM) is a primary Ca2+ signal transducer protein that undergoes multiple conformations upon Ca2+ binding, which allow it to bind and activate a myriad of target proteins. An appropriate force field to model the effect of Ca2+ on the dynamics of CaM is still lacking. Here, we present a coarse-grained approach using the Associative memory, Water mediated, Structure and Energy Model (AWSEM) force field to study the Ca2+-induced conformational changes on CaM. We have parameterized the force field for the Ca2+-CaM to match the experimental observables. This model provides mechanistic insights on CaM's ability of target binding and selection.

Abstract N° 94

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Central Nervous System and serotonin provide distinct guidance cues for hindgut motility

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Intestinal motility is a critical motor function of the gastrointestinal (GI) system governed by multiple neurogenic systems. Due to its major role in maintaining homeostasis, several overlapping mechanisms have evolved for its adaptive operation including modulation by the Central Nervous System (CNS), Enteric Nervous System (ENS), and gut pacemaker cells . Deficits in intestinal motility are poorly understood due to the complexity of neurogenic motor patterns and their inaccessibility for direct observations in traditional mammalian models. Crayfish offer a unique and accessible model to study the GI tract ex vivo and as well as understand the role of the CNS and neurohormones in regulating motor control mechanisms. In this study, we investigated the effects of denervation and serotonergic excitation on crayfish hindgut motility. Serotonin was found to provide the required power to move muscles outlining the gut whereas CNS maintained coordination of gut motility in space and time.

Abstract N° 95

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Double labeling strategy using non-canonical amino acids for accurate single molecule FRET measurements on a G protein coupled receptor

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Metabotropic glutamate receptors (mGluRs) are class C G-protein coupled receptors (RPCG) essential for the fine tuning of the glutamate response in synapses of the central nervous system. Upon activation by glutamate, binding on the extracellular domain, these dimeric receptors undergo a multidomain conformational reorganization that allows for G-protein interaction. Understanding these conformational changes is key for the design and discovery of innovative drugs.

Here we use genetic code expansion to introduce two distinct non-canonical amino acids in response to the introduction of two different stop codons in mammalian cells. This allows us to control the positioning of donor and acceptor fluorophores to produce orthogonally labeled receptors for single molecule FRET measurements. Single molecule FRET changes in presence of different ligands give information about the conformational changes that the receptors undergo, while lifetime measurements uncover the dynamic of these changes (1). Our double labeling strategy has the potential to accurately account for photophysical changes of the fluorophores upon attachment to the receptor to perform accurate FRET and distance measurements (2).

Measuring accurate distances by single molecule FRET will be central for the characterization and modeling of the conformational transformations as well as intermediate states explored by mGluRs and induced by both orthosteric and allosteric ligands (1).

Abstract N° 96

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Universal biofilm height dynamics

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Bacterial colonies play important roles in ecology and medicine, as they are sources of infection and they account for a large percentage of biomass on Earth. However, fundamental questions remain about the nature and dynamics of the vertical growth of biofilms. Through interferometry, we can capture their height dynamics with nanometer precision. With this tool, we characterize their growth and show that current literature models do not capture the behavior of the system on short and long timescales.

Using a data focused approach we develop a simple model that takes in account the geometry of the system, and the fact that the interaction of the colony and the environment is limited through its interface. This interface-limited model, accurately fits the growth of multiple strains in a laboratory setting (e.g., E. coli, V. cholerae, Aeromonas, Bacillus, yeast) for the first 48 hours, and predicts the maximum height a colony reaches in the long time limit. While this interface-limited model accurately describes height dynamics across time scales, more experiments and analysis must be done to understand the limits in which it remains useful

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Abstract N° 97

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Transcription-limited growth

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Most of the current quantitative approaches to biosynthesis are

translation-centric. Yet, transcription is an essential step of protein

synthesis and therefore for cellular growth, as revealed by a number of

important phenotypes such as aneuploidy and senescence. Recent attempts

addressing role of transcriptional limitations have focused on the

growth mode of single cells (linear vs exponential), but many other

questions remain open. In this work, we develop a theoretical resource

allocation framework for biosynthesis including transcription, and we

explore broadly the consequences of transcriptional limitations on

known growth laws, showing that they apply to both bacteria (E. coli)

and eukaryotes (budding yeast). We first consider the effect of

transcription limitations on the well-known problem of the growth cost

of useless protein production. If the burden were purely due to

translation, the cost of a useless protein would decrease as the

stability of its mRNAs decreases (vanishing in the ideal case of

completely unstable mRNAs). However, this is not the case, as data from

the Barkai group (Kafri et al., Cell Rep 2015) clearly falsify this

scenario in budding yeast. Our quantitative model of biosynthesis

correctly predicts the experimentally measured growth cost of a useless

gene as a function of both gene copy number and mRNA stability. In

particular, the cost does not scale with the concentration of the

useless protein if transcription is limiting growth. Analyzing data on

E. coli, we are able to show that growth is increasingly limited by

transcription in poorer nutrients. In accordance with this fact, our

model predicts that for growth-optimized resource allocation both total

mRNA concentration and the fraction of ribosomes bound to mRNA decrease

as nutrient conditions worsen. As a consequence of this limitation,

transcription-targeting drugs are predicted to be more effective in

richer media. These findings clear the way for more specific

explorations of physiologically relevant situations where growing or

growth-arrested cells are limited by transcription.

Abstract N° 98

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Probing T cell thymic maturation through repertoire sequencing

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In order to protect themselves against external threats, vertebrates have developed an adaptive mechanism of immune defence where a leading role is played by T lymphocytes, cells able to identify short protein fragments of viral origin through a unique receptor. After the stochastic generation of their receptor, T cells undergo a selection process known as thymic maturation which aims at releasing into the host periphery only those

cells with functional and weakly self-reactive receptors. From high-throughput sequencing of different provided repertoires FACS-sorted in the thymus, we use inference techniques to learn interpretable models for the selected features, informative about the maturation cascade. Our results suggest that thymic maturation is a stochastic process of single cell promotion enhanced by cell-cell interactions.

Abstract N° 99

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Physical modeling of ribosomes along messenger RNA with a ballistic model: application to the estimation of the kinetic parameters from ribosome sequencing experiments

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Genetic translation is a crucial stage for cells as they need proteins to carry out vital functions. From the time of the discovery of messenger RNA, biophysicists have attempted to model the transport phenomena behind the translation of a gene by ribosomes. In the 2010's a new type of experiment called Ribo-Seq was developed and provided a substantial contribution to the study of translation. With Ribo-Seq data, biophysicists are trying to apply and improve previous translation modeling approaches. It turns out that the classical Ribo-Seq experiments do not provide sufficient information to deduce kinetic ribosome hopping rates. To overcome this, we have developed a novel experimental setup consisting in separating the messenger RNAs into categories that depend on the number, from one to four, of ribosomes. We model the motion of ribosomes along mRNA with a ballistic model where particles advance along a filament without excluded volume interactions. Our model also incorporates the degradation of mRNA. The additional equations coming from these different Ribo-Seq profiles allow us to fit the initiation rate and all the hopping rates.

Abstract N° 100

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Study of the mechanical impacts on the phenotypic plasticity of Candida albicans in vitro using microfluidics.

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The filamentous yeast Candida albicans is a benign member of the healthy human microbiota but can also turn into an opportunistic pathogen in immunocompromised patients. The morphological plasticity of C. albicans is a major virulence factor. Hyphae, i.e. elongated multi-cellular invasive filaments, and yeasts respectively participate to invasion and dissemination in the tissues and bloodstream. The reversible yeast-to-hypha phenotypic switches depend on the physicochemical properties of C. albicans environment, notably pH, temperature and growth factors. C. albicans also secretes farnesol, a quorum-sensing molecule that is involved in inhibiting hyphal induction.

In contrast to the impact of chemical parameters, little is known about how mechanical forces affect the growth phenotype. To this end, we have developed dedicated microfluidic devices allowing both live-cell imaging and high-throughput experiments to quantify the impact of

compressive stress on the yeast-to-hypha transition. In addition, the compressive chamber was connected to a second stress-free compartment through narrow channels only accessible to filaments, to study hyphae mechanical relaxation.

Using these devices, we observed expression the hyphal-specific gene HWP1 under very small compressive stress intensities, compatible with physiological values observed in human host (dozens of kPa). Yeast-hypha transition probability in the confined population seems modulated by the concentration of secreted farnesol and cell-division rate. Single-cell reporter expression also showed that higher compressive stress values (hundreds of kPa) limits hyphal-growth maintenance. In contrast, observation of hyphae in the second uncompressed chamber showed that most hyphae produced lateral yeasts while maintaining polarized tip growth, suggesting some degree of reversibility in the process.

Together, our results call for a better apprehension on how mechanical stress and chemical cues couple to drive the dynamics of C. albicans phenotypic plasticity. A better understanding of these biophysical effects is critical to have a deeper insight of the adaptation of this microorganism to the various mechano-chemical environments of the host.

Abstract N° 101

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Dynamic Changes in the Replication Complex During Normal and Perturbed Replication

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The mechanisms of DNA synthesis are closely inspected throughout the entire cell cycle. The licensing of chromatin, the initiation, elongation and termination of replication are subjected to specific control by different mechanisms. During unperturbed replication these mechanisms monitor for balanced distribution of exact DNA copy number in prodigy cells. When replication is compromised, the S-phase checkpoint is activated in order to stabilize the replication complex and to provoke the repair response for damage processing. In order to obtain a complex picture of the entire process of DNA replication, the specific characteristics of the dynamics of interactions in-between the replisome proteins must be obtained and evaluated. That is why, by means of highly sensitive and precisely aligned microscopy system, we study the changes in the dynamics of different replication-associated proteins throughout replication. Our live-cell imaging approach permits precise detection of process fluctuations that are in a range of microseconds. It aims to obtain quick and user-friendly model for genome instability diagnostics and drug-testing.

Abstract N° 102

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Developing a 3D spheroid model to study how tumor-associated macrophages evolve in the early melanoma tumor microenvironment

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Melanoma, the deadliest form of skin cancer, affects over 100,000 people each year in the United States alone. Treatment with immune checkpoint inhibitors (CPIs) aimed at boosting T cell function have found some success in melanoma patients; however, 50% of patients do not have objective responses. Tumors unresponsive to CPIs often display increased numbers of tumor-associated macrophages (TAMs), which are generally immunosuppressive and associated with a poorer prognosis, making TAMs a promising therapeutic target.

The immune activity of TAMs and other cells in the melanoma tumor microenvironment (TME) is determined by cell-cell interactions and a network of secreted signals. We have previously shown that intercellular signaling networks between macrophages, fibroblasts, and melanoma cells are central to establishing and maintaining an immunosuppressive TAM phenotype. Understanding how TAM subpopulations are recruited and evolve within the tumor and their functional role in the immunosuppressive signaling network will support development of targeted therapies to reestablish an effective immune response, especially in cases of CPI resistance.

To elucidate this network, we constructed a 3D in vitro spheroid model of the melanoma TME to study interactions between a murine melanoma tumor cell line (YUMMER1.7), bone marrow derived macrophages, and fibroblasts (3T3MEF). Spheroid models have been shown to best recapitulate the physical properties of tumors, including direct cell-cell interactions and spatial orientation of cells. Here, we demonstrate the ability to form spheroids with these three cell types embedded into a 1.5mg/ml Collagen I gel, which can be imaged over 7 days to capture spheroid growth and evolution.

In subsequent studies, we aim to track macrophage subpopulations and how they are enriched or excluded from the spheroid core, and we will analyze the phenotypes of these macrophages using immunofluorescence staining. Additionally, we will use this model to study recruitment and infiltration of TAMs by embedding a spheroid within a suspension of macrophages. These ongoing studies provide a framework to understanding the contribution of macrophages in the evolution of the immunosuppressive phenotype in early melanoma.

Abstract N° 103

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Investigation of the conformational dynamics of chaperone proteins using single molecule FRET

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Newly translated amino acid chains have to fold into their native structures in order to perform their vital functions. Understanding the rules governing the correct folding of protein domains is of high interest in biochemical research. The heat shock protein (Hsp) family assists the successful folding of proteins to their native structures. They are also responsible for preserving protein structure in case of disruptive stimuli such as elevated

temperatures. The Hsp proteins have been studied for decades, however their underlying mechanism of action is still controversial and under intense investigation. Meanwhile, single molecule techniques have become powerful approaches to reveal the conformational dynamics of proteins. Single molecule experiments can resolve coordinated motions of proteins in great detail due to their sensitivity and ability to monitor dynamic changes in real-time. The aim of this work is to study the dynamics of various heat shock protein systems with single molecule FRET experiments. We have conducted 2-color and 3-color single molecule FRET experiments to study the conformational dynamics of several chaperone systems including Hsp70, Hsp90 and their co-chaperones.

Abstract N° 104

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Hyphal network and Brownian motion

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The success of filamentous fungi in colonizing most natural environments can be largely attributed to their ability to form an expanding interconnected network, the mycelium, or thallus, constituted by a collection of hyphal apexes in motion producing hyphae and subject to branching and fusion. In this work, we characterize the hyphal network expansion and the structure of the filamentous fungus Podospora anserina under controlled culture conditions. To this end, temporal series of pictures of the network dynamics are produced, starting from germinating ascospores and ending when the network reaches a few centimeters width, with a typical image resolution of several micrometers. The completely automated image reconstruction steps allow a post-processing and a quantitative analysis of the spatio-temporal dynamics. With this information, each hypha can independently be seen as a trajectory of a particle subject to drift and diffusion; a realisation of a system following the Fokker-Planck equation. We aim to describe the thallus as a collection of hyphae trajectories that can be described with only 2 parameters derived from the Fokker-Planck equation. This will allow to predict the probability of hyphae density at a given position and a given time.

Abstract N° 105

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The role of DNA strand displacement kinetics in CRISPR-Cas r-loop formation

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DNA strand displacement is a fundamental process by which a single-stranded nucleic acid (either DNA or RNA) invades a duplex, annealing to a complementary substrate strand and displacing the incumbent strand. Strand displacement plays a role in several biological processes such as homologous recombination, the resolution of the transcription bubble, and the CRISPR-Cas system. In the CRISPR-Cas system, sequence specificity is accomplished using a guide RNA sequence that invades a DNA duplex and forms an R-loop. This is the

rate-limiting step in Cas-directed DNA cleavage, suggesting that DNA strand displacement kinetics may play a significant role in overall Cas cleavage rates. Here, we present a single-molecule technique for studying the displacement time of DNA strand displacement events. Using this technique, we demonstrate that displacement rates depend strongly on the sequence of the invasion domain for short (14bp) sequences. We present results which suggest that DNA and RNA invaders invade duplexes at different rates, though neither nucleic acid appears to be uniformly faster. We also demonstrate that the side of invasion matters; invasion in the 5'-to-3' direction occurs at different rates than invasion in the 3'-to-5' direction, even when the sequence of base pairs being broken is identical. This result is of particular interest because Cas proteins invade in specific directions: Cas12 displaces the incumbent starting from the 5' end, while Cas9 begins at the 3' end. Finally, we describe plans for a systematic study of the sequence dependence of DNA strand displacement which can be extended to direct study of Cas protein R-loop formation.

Abstract N° 106

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Structure and mechanics of the human Nuclear Pore Complex basket

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Nuclear Pore Complexes (NPCs) are huge protein assemblies, embedded in the nuclear envelope to ensure the controlled nucleo-cytoplasmic trafficking of proteins and mRNAs. While decades of investigation have provided a detailed view of the molecular arrangement of the central scaffold of NPCs, very little is known about the peripheral regions. We investigated the structure and mechanical properties of the nucleoplasmic side of NPCs, designated as "basket" after its fishtrap-like architecture. To do so, we implemented a correlative Atomic Force Microscopy / fluorescence microscopy approach. We show that the basket has a highly variable structure and can explore a large volume within the central NPC channel. Finally, we investigated the mechanical properties of NPCs and found that the basket filaments are softer than the NPC scaffold, which corroborates its structural flexibility.

Abstract N° 107

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Characterizing Nuclear Deformation in Activated T-Cells

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T-cells, a key component of the adaptive immune system, are activated when receptors on their surface recognize and bind to pathogenic peptides on antigen-presenting cells. The contact region between the two cells is known as the immune synapse, and its formation is a critical step in the adaptive immune response to pathogens. T-cell activation is a dynamic process resulting in dramatic changes in cellular morphology and cytoskeletal organization. Prior work has established that T-cell activation leads to chromatin reorganization and gene repositioning, but the mechanisms by which activation can alter nuclear shape and mechanics in T-cells remain largely unexplored experimentally. Here we use confocal microscopy to study nuclear deformation in T-cells activated on a planar glass surface. We have developed an analysis platform to characterize global and local measures of nuclear shape and deformation as a function of external stimuli. The platform generates 3D tetrahedral mesh reconstructions from volumetric data and computes curvatures at each face of the mesh. From these curvature calculations, areas of the nucleus surface are classified as concave or convex, with the concave regions corresponding to grooves on the surface. The centrosome is found to be consistently situated in the vicinity of a deep groove, with this groove descending towards the plane of the immune synapse upon activation. We find that while the volume and surface area of the nucleus do not change appreciably upon T cell activation, the nucleus deforms significantly and spreads out, with grooves occupying a larger fraction of its surface area. We also introduce mechanical perturbations to the system to investigate the determinants of nuclear morphology. Treatment with DZNep, an inhibitor which softens the nucleus, renders the nucleus more pliant to external forces applied by the cytoskeleton. We find that DZNep treatment does not affect the volume, surface area, or prevalence of grooves on the surface but enhances the nuclear spreading already observed upon activation.

Abstract N° 108

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Interdependence between mRNA translation and growth : the cost of unnecessary protein synthesis

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The amount of RNA and proteins in bacteria depends on the cellular growth rate, which is known as the bacterial growth law. The ribosomal mass fraction φ is known to increase linearly with the growth rate Λ, when nutrient conditions are the limiting step for growth. If translation inhibition is induced, a second law describes the (linear) increase of φ when Λ is decreased by sub-lethal doses of antibiotics.

We develop a theoretical formalism describing the competition of mRNAs for ribosomes to recover the phenomenological growth laws. Traffic models such as the TASEP (totally asymmetric simple exclusion process) are used in a mean-field approximation, and the effect of unnecessary protein synthesis on the cellular growth rate is analysed.

Abstract N° 109

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Identifying phenotypes associated with coupling the epithelial to mesenchymal transition and metabolic reprogramming

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The majority of cancer related deaths are caused by metastasis. To determine which pathways to target for treatment a complete picture of tumorigenesis, growth, proliferation and metastasis is needed. The epithelial to mesenchymal transition (EMT) is critical to metastasis as a hybrid epithelial/mesenchymal (E/M) phenotype allows for the collective migration of cells. Additionally, cancer cells are known to consume large amounts of energy through the Warburg effect (aerobic glycolysis). Although cells under normoxic conditions use oxidative phosphorylation (OXPHOS), previous results have shown cancer cells can have a hybrid metabolic phenotype by using both aerobic glycolysis and OXPHOS (W/O). While recent studies have provided insight into EMT and metabolic reprogramming independently, experimental results indicate transcriptional and miRNA regulatory coupling between the two networks. Our approach couples the core circuit of EMT with the core metabolic circuit through a mechanism based computational model. The model includes five transcriptional regulations of the EMT circuit by the metabolic network, three miRNA regulations of the metabolic circuit, and one external input to the EMT circuit. Initial results show an upregulation of the coupled hybrid/hybrid (E/M-W/O) phenotype when either the E/M phenotype or hybrid metabolic phenotype is upregulated. Additionally, strong regulation between the core circuits can suppress all coupled phenotypes except the hybrid/hybrid (E/M-W/O). These results suggest the aggressive metastatic phenotype (E/M) requires an excess of energy (W/O) and propose initial therapeutic targets to disrupt the expression of the cancer phenotypes.

Abstract N° 110

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Dynamics of Chromosome Organization During the Cell Cycle of a Minimal Bacterial Cell

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The minimal bacterial cell, JCVI-syn3A, is the result of synthetic genome minimization cycles, ending with a 543 kbp genome with 493 genes. Genome-scale gene essentiality and proteomics, along with the essential metabolism of Syn3A were previously determined. Whole-cell models (WCMs) of Syn3A have been developed by combining a kinetic model of the metabolism with a model of genetic information processing. In previous work, we reconstructed cell geometries of Syn3A, including the ribosome distribution,

from cryo-electron tomography and generated ensembles of self-avoiding circular chromosome configurations on a lattice, which were constrained by the cell geometry and ribosomes. Initially, these configurations of the DNA were static within any individual simulations of the WCM and their heterogeneous crowding influenced the diffusion of biological macromolecules, such as RNA polymerases (RNAP) and mRNA, throughout the course of the simulations. A continuum model of Syn3A's chromosome is now being developed to model the dynamics of the chromosome's organization over the course of a complete cell cycle, which will include both DNA replication and cell division. Within the continuum model, the DNA is modeled as a stiff homopolymer with a torsional stiffness at a resolution of 10 bp per monomer. A hybrid methodology is being developed to couple the spatially-resolved WCM to a Brownian dynamics simulation of the DNA polymer and ribosomes. This allows for changes in the accessibility of genes to RNAPs, kinetics-based replication to govern the processive motion of replication forks, and segregation of daughter chromosomes during cell division. As was confirmed by 3C-seq, Syn3A lacks systems governing chromosome organization, leaving the fundamental processes common to all bacteria as the sole factors influencing chromosome organization.

Abstract N° 111

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Abstract N° 112

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Local Epigenetic Landscape Controls the Integrity of Topologically Associating Domains upon Cohesin loss

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The 3D organization of chromosomes is intertwined with gene regulation. Chromatin looping and the segregation into active and inactive compartments are widespread principles of genome folding. Many depletion studies have reported that CTCF and cohesin complex play a crucial role in genome folding by mediating chromatin loop. A key experimental finding is that cohesin deletion abolishes both cohesin-associated CTCF loops and TADs, while the compartment structure is maintained and even enhanced. However, using our chromosome copolymer model (CCM), here we show that a subset of TADs is conserved after removal of CTCF loop. Interestingly, the preservation of some TADs upon CTCF loop loss can be explained by the underlying epigenetic state. The TADs whose boundaries align with the switches in active/inactive state signals are retained despite CTCF loop deletion, which is also supported by our analysis on Hi-C contact matrices in the presence/absence of cohesin. Consequently, our results suggest that the underlying epigenetic landscape is responsible for controlling

mammalian chromatin organization upon CTCF loop deletion not only at the level of compartments but also at the level of TADs.

Abstract N° 113

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Friend or Foe: The Potential of Transposable Element Tuning

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Transposable elements (TEs) are mobile DNA sequences that can be cut or copied and reintegrated elsewhere into the genome. Long considered junk DNA, TEs have instead been determined to be major contributors to disease, development, and evolution. The evolutionary arms race between TEs and the DNA modification systems that curb the damage of TE proliferation has created a delicate balance of TE regulation in our cells. However, this tight control of TE propagation loosens over the course of our lives. The deregulation of TEs has been associated with the onset of several developmental diseases and cancers. On the other hand, increased TE activity in cancerous cells in blind mole rats has been shown to impart them with cancer resistance by inducing concerted cell death of diseased cells. The difference in these contrary outcomes is the tuning of TE activity, demonstrating the importance of characterizing TE dynamics. The potential benefits are twofold: in the prevention of TE-associated diseases and the promise of developing TEs as tools for gene regulation or therapeutic purposes. In this study we characterize the dynamics of the autonomous TE, IS608. Several questions about IS608's transposition process have remained unanswered. Firstly, IS608 transposition may or may not result in TE replication. Secondly, IS608 contains a gene of unknown function, TnpB. We address these questions through real-time tracking of fluorescence-tagged genetic sequences, that allow us to measure transposase, TnpB protein and excision levels in live cells. We show that TnpB works with transposase to increase the active TE count, not through replication but through conservation of active TEs. We propose a model in which TnpB increases the efficiency of successful reinsertion of excised TEs. TnpB may achieve this by use of its RuvC-like domain to improve TE insertion.

Abstract N° 114

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Traffic slowdown by antibiotics

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The mRNA translation process is often described by a non-equilibrium traffic model called the Totally Asymmetric Simple Exclusion Process (TASEP), in which ribosomes are allowed to move uni-directionally on a one-dimensional lattice (the mRNA), and the particle current determines the overall protein synthesis rate. Similarly to what it has been done in the past in the context of RNA polymerases, in this work I want to extend the canonical model by introducing a pausing state of the ribosome and analyse its effects on the ribosomal current and ribosome density.

This model mimics the impact of antibiotics affecting the elongation cycle of the ribosome, and allows for a quantification of the amount of actively translating and pausing ribosomes. I will couple this system to a reservoir of free ribosomes, and observe how they are allocated when different antibiotic concentrations are added into the system.

Abstract N° 115

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High Throughput Platform for Imaging Worms' Microbiome Composition Dynamics

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With the growing number of implications of the microbiome in health and disease, there is a growing interest in understanding the relationship between microbial composition and host health. This composition has been found to be highly dynamic and idiosyncratic, influenced by the environment, host-microbe interactions and microbe-microbe interactions. To interrogate these factors and build a model of the microbiome, it is necessary to track compositional dynamics in the context of a host in a controlled environment. This is difficult to do in mammalian hosts, but could be done in small model organisms.

Here I describe an experimental system that uses the nematode C. elegans and its functionally-representative 10-species consortia of microbes as a model system. This model organism has the advantages that it's been extensively studied and has a short lifespan, transparent body, and an extensive genetic library/toolkit. Its microbiome has also previously been studied with deep sequencing, metabolic reconstruction and genetic tools available. The large fluctuations in this system, and the combinatorial explosion of microbes necessitate an experimental setup of significant throughput. To this end, we developed a multiplexed microfluidic platform for spatiotemporal imaging of bacteria in the gut of C. elegans. By feeding fluorescently labeled bacteria to worms confined in a microfluidic device placed on a fluorescence microscope it is possible to spatially and longitudinally track the microbial composition in the gut of single animals in real time over the course of many hours. This system is designed to generate quantitative dynamical data for development of generalized Lotka-Volterra models to link the compositional dynamics with interspecies interactions and host physiology. Real-time control over the host environment is designed to test predictions made by these models.

The key design requirements of this setup include high throughput, combinatorial and temporal control of host microenvironment, durability in the face of bacterial adhesion, fully automated multichannel imaging at high resolution and fast processing on the fly. I will describe a combination of engineering and molecular biology approaches to address these requirements, and present preliminary studies on the dynamics and control of microbiome composition.

Abstract N° 116

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Fundamental behaviors emerge from simulations of a living minimal cell

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We present a whole-cell fully dynamical kinetic model (DWCM) of JCVI-syn3A, a minimal cell with a reduced genome of 493 genes that has retained few regulatory proteins or small RNAs. Cryo-electron tomograms provide the cell geometry and ribosome distributions. Time-dependent behaviors of concentrations and reaction fluxes from stochastic-deterministic simulations over a cell cycle reveal how the cell balances demands of its metabolism, genetic information processes, and growth, and offer insight into the principles of life for this minimal cell. The energy economy of each process including active transport of amino acids, nucleosides, and ions is analyzed. DWCM reveals how emergent imbalances lead to slowdowns in the rates of transcription and translation. Integration of experimental data is critical in building a kinetic model from which emerges a genome-wide distribution ofmRNA half-lives, multiple DNA replication events that can be compared to qPCR results, and the experimentally observed doubling behavior.

Abstract N° 117

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Tuning to criticality in the cochlea

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The human ear can detect sounds over a vast frequency range with incredible sensitivity. To explain these properties, it has been suggested that individual hair cells operate near an oscillatory instability, a so-called Hopf bifurcation. However, resonating modes are determined primarily by the passive hydrodynamics and mechanical stiffness of the cochlea, with active processes in individual hair cells only serving to counteract dissipative forces. In this picture, every frequency must have a mode tuned close to a Hopf bifurcation, but each hair cell impacts every mode to varying extents. It is an open question how these coupled hair cells could tune their behavior so as to bring the entire cochlea to this line of Hopf bifurcations. To approach this question, we develop a model where hair cells collectively tune their active processes while only using information measurable from their location on the cochlea. Our work suggests a new perspective on the hearing mechanism: Instead of individual hair cells being critical, their concerted individual action tunes the whole cochlea to a critical point revealing a collective organization principle in the inner ear.

Abstract N° 118

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Chromosome Modeling on Downsampled Hi-C Maps

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The human genome is organized within a nucleus where chromosomes fold into an ensemble of different conformations. Chromosome conformation capture techniques such as Hi-C provide information about the genome architecture by creating a 2D heatmap. Initially, Hi-C maps experiments were performed in human interphase cell lines. Recently, efforts were expanded to several different organisms, cell lines, tissues, and cell cycle phases where obtaining high-quality maps is challenging. Poor sampled Hi-C maps present high sparse matrices where compartments located far from the main diagonal are difficult to observe. Aided by recently developed models for chromatin folding and structure, we develop a framework to enhance the compartments' information far from the diagonal observed in experimental sparse matrices. The simulations were performed using the Open-MiChroM platform aided by new trained parameters into the Minimal Chromatin Model (MiChroM) energy function. The simulations optimized on a downsampled experimental map (10% of the original data) allow the prediction of a similar contact frequency to the complete (100%) experimental Hi-C. The modeling results open a discussion on how simulations and modeling can increase the statistics and help fill in some Hi-C regions not captured by poor sampling experiments. Open-MiChroM simulations allow us to explore the 3D genome organization of different organisms, cell lines, and cell phases that often do not produce high-quality Hi-C maps.

Abstract N° 119

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Brain Imaging of a Freely Moving C. elegans performing thermotaxis

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The nematode C. elegans shows a thermotactic behavior where it remembers the temperature it has been habituated to. Worms habituated to a certain temperature Tc for more than 4 hours tries to return to Tc when put on a temperature gradient.

The negative thermotaxis behavior of a worm placed in T>Tc is in general believed to be more robust than the positive thermotaxis behavior of worms at T<Tc. The temperature range where we can observe this behavior is larger and some mutants only perform negative thermotaxis.

In this work, we perform calcium imagine of multiple neurons in a C. elegans performing thermotaxis. We use a spinning disk confocal microscope equipped with a thermal gradient stage to perform volumetric imaging of a freely moving C. elegans. We record the calcium intensity from the interneurons AIY,AVA,RIM and the motor neurons RME,SMD, while the worm freely moves around the thermal gradient.

We discuss preliminary signals we find in the neural state related to the behavior of the worm.

Abstract N° 120

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Exploring the hierarchical structural organization of squid iridocyte protein

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Squids of the family loliginidae are primarily known for displaying a wide array of colors and instantaneous color patterns, called iridescence, to camouflage as a mechanism to blend into its surroundings. Reversible self-assembly of reflectin into nanoscale (50-100 nm) protein platelets that gets wrapped by the lipid membrane and sandwiched between the extracellular matrix provides a differential refractive index interface for light to reflect off and undergo constructive interference in the squid iridocytes. Depending on the shape, thickness and arrangement of these protein platelets, different optical effects emerge and by tuning the reversible assembly of reflectin, squids exhibit a virtually unlimited palette of colors. The key to understanding the supramolecular assembly of reflectin requires characterization of the hierarchical structural organization of the protein from the primary sequence to the quaternary structure. Reflectins are disordered proteins with an unusual sequence composition enriched in aromatic and positively charged amino acids, with multiple repeats of a conserved domain, and unusually devoid of small nonpolar residues necessary for hydrophobic collapse. We have identified that reflectin exists in a polydisperse state, and the oligomeric complexes have a highly positively charged surface with a predominant beta sheet secondary structure. We hypothesize a patchy particle self-assembly model to be involved in the formation of reflectin platelets. Anisotropic patterns on the protein surface or shape that could act as enthalpic or entropic patches respectively could explain the sticking of reflectins to each other to form the high-index solid platelet with a low packing fraction. We use a set of biophysical and biochemical techniques to pin down the source of this stickiness and use different imaging tools to uncover the dynamics of this macroscopic assembly.

Abstract N° 121

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The SuperBeeLive macroscope, a monitoring tool to investigate the bee-environment relationship.

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A combination of environmental stresses including chemical pollution, invasive predatory or parasitic species, or climate change is producing, on a global scale, an undeniable decline in bee colonies. This damage to the main pollinating species is not only a precursor of important losses in agricultural production but also a testimony to the degradation of the Earth's ecosystem. Therefore, the study of bee health must be considered

from a holistic perspective and mobilize the concerted efforts of distinct communities of knowledge and expertise.

The "SuperBeeLive" project that we are conducting consists in analyzing in real time the social organization and homeostasis of a bee colony, from birth to death, while monitoring the levels of stress intensity that it undergoes through several biological interactions and exposure to neurotoxic insecticides. The innovation consists in the construction of a macroscope for the observation of bee colonies thanks to the instrumentation of a beehive in order to continuously collect several types of digital and video data streams in parallel, in a systematic, automatic and non-invasive way. This experimental device, which uses IoT, computer vision and neural network technologies, is intended (i) to decipher the etiology of the bee's disorders by correlating observed behavioral colony disturbances with the variations of these stressors (ii) to increase our knowledge of the functioning of the bee superorganism.

Abstract N° 122

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Radius of gyration scaling of proteins compared to polymer model simulations

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The radius of gyration (Rg) has long been used as a metric for characterizing the average dimensions of a protein. Similar to Rg, one can calculate the average dimension of each consecutive protein segment n of the overall protein length N to observe how the radius of gyration scales with increasing n, known as the Rg(n). The Rg(n) is described by the equation: Rg(n) ≈ An^v, where A is a coefficient, n is the length of the segment, and v is the scaling exponent. We found when observing the Rg(n) curves for the Dunbrack 1.8 dataset that proteins have two distinct regimes where power-law scaling of the Rg(n) is present: one where n is below 20, and one where n is above 20. The power-law scaling rate for each of these regions v is ~0.68 and ~0.24 for n < 20 and n > 20, respectively. We propose the n < 20 region has a higher v due to secondary protein structure elongating shorter length protein segments, while the n > 20 region has a smaller v due to the folded, tertiary structure of most proteins. Notably, these v values are different than what is expected for the growth rate of a sticky polymer according to Flory theory (v = 0.33). In this report, we observe the Rg(n) scaling of proteins and compare it to several polymer simulations, including random walk polymers, jammed polymers, and sticky polymers. In doing so, we show how protein secondary and tertiary organization contributes to the differences in protein Rg(n) scaling rates from these simulated systems.

Abstract N° 123

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Transcription factor dynamics in gene expression: the long and short of it

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Transcription factors (TFs) regulate gene expression by binding to specific consensus motifs within the local chromatin context. The mechanisms by which TFs navigate the nuclear environment as they search for binding sites remain unclear. Single-molecule tracking (SMT) has emerged as a powerful approach to explore mechanisms of transcription factor movement and chromatin interactions in living cells. We used single-molecule tracking and machine-learning based classification to directly measure the nuclear mobility of a broad spectrum of transcription factors in live cells (including the glucocorticoid receptor and the estrogen receptor). We found that TFs follow power-law distributed binding times, suggesting that productive binding to initiate transcription occurs over a broad range of timescales and may involve longer binding events than previously thought.

Analysis of single molecule trajectories of the glucocorticoid receptor revealed two distinct and dynamic low-mobility populations. One population represents specific binding to chromatin, while the other represents a confinement state that requires an intrinsically disordered region (IDR). The dwell times of both populations follow a power-law distribution, consistent with a broad distribution of affinities of GR/chromatin binding and interactions. Our data link IDRs with a confinement state that is functionally distinct from specific chromatin binding and modulates the transcriptional output by increasing the local concentration of TFs at specific sites. Our results imply that a potential interaction between the heterogeneity of the nuclear microenvironment and intrinsic conformational dynamics defines the specific and non-specific chromatin binding of transcription factors, thereby regulating transcription.

Abstract N° 124

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Multi-omics topic modeling for breast cancer classification

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Topic modeling is a widely used approach to extracting relevant information from large datasets. Recently the problem of finding a latent structure in a dataset to the community detection problem in network theory and a new class of topic modeling strategies has been introduced to overcome some of the limitations of classical methods. We tested this approach on lung and breast cancer samples from the TCGA and METABRIC databases, using data of messenger RNA, microRNAs, and copy number variations. The established cancer subtype organization is well reconstructed in the inferred latent topic structure. Moreover, the "topics" that the algorithm extracts correspond to genes involved in cancer development, and they are enriched in genes known to

play a role in the corresponding disease; they are strongly related to the survival probability of patients too.

In biology, integrating transcriptional data with other layers of information, such as the post-transcriptional regulation mediated by microRNAs, can be crucial in identifying the driver genes and the subtypes of complex and heterogeneous diseases such as cancer. More specifically, we show how an algorithm based on a hierarchical version of stochastic block modeling can be adapted to integrate any combination of data. We will also show that the inclusion of the microRNAs layer significantly improves the accuracy of subtype classification. As a final result, we show how operating in the low dimensional topic space one can predict, the cancer subtype of a test expression sample.

Abstract N° 125

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Emergent property of collective migration: spatial organization of phenotypes

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To invade new territories, groups of bacteria consume and collectively chase attractant cues using chemotaxis. This strategy provides strong fitness advantages for a population to search for nutrients and grow rapidly. Though decades of quantitative studies have elucidated the dynamic of bacterial migration, how non-genetic diversity contributes to this dynamic is not fully understood. Collective migration requires coordination, but cells within an isogenic population exhibit diverse motility behaviors. Previous work in our lab demonstrated that how fast bacterial cells climb chemical gradient, or chemotactic performances, depends on their swimming phenotypes and the environments. As a result, cells are spatially organized based on their chemotactic performances during collective migration; cells with higher performances leads at the front, where the attractant gradient is low, and cells with lower performances stayed at the back, where the attractant gradient is high. While this was demonstrated in liquid, the same principle predicts that spatial organization of phenotypes is different in other environments. Here, we tested this prediction by measuring organizations of phenotypes in cell populations that are migrating in porous environments. Our results will demonstrate that populations can leverage non-genetic diversity to migrate through multiple environments by altering their leader-follower structures.

Abstract N° 126

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CONFORMATIONAL ENSEMBLE OF RNA IN BIOMOLECULAR CONDENSATES USING ATOMISTIC MOLECULAR DYNAMICS SIMULATIONS

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Biomolecular condensates formed through liquid-liquid phase separation (LLPS) of proteins and RNAs are nowadays accepted to play a key role in cellular organization. Despite the ever-growing attention in

cellular biology, a thorough characterization of the detailed molecular mechanism underlying LLPS process is still not fully achieved.

Recently, our group has introduced a “divide-and-conquer” strategy to investigate the microscopic origin of LLPS by accurate atomistic molecular dynamics simulations of concentrated mixtures of small peptides/nucleotides. This approach has been used to disentangle the molecular interactions driving the LLPS of DDX4 proteins [1] and to investigate heterotypic and homotypic interactions in condensates formed by polyR/polyK peptides with polyU [2].

Building on these findings, we are currently extending this analysis to structured RNA fragments in order to shed light on their conformational ensemble in biomolecular condensates. In particular, we focused on a previously well-characterized RNA hairpin [3] in the presence of arginine/lysine-rich polypeptides at a concentration comparable to the biomolecular density in cellular condensate.

To this aim, we performed explicit solvent all-atom molecular dynamics and to tackle sampling problems we took advantage of enhanced sampling techniques, combining parallel tempering with metadynamics.

In our first results we observed that polypeptides significantly interact with the RNA hairpin influencing its conformational dynamics. Furthermore, we noticed a differential behavior based on the characteristic sequence of the peptide fragments, in particular arginine-rich polypeptides are able to clearly reshape the folding landscape of the hairpin.

[1] Paloni et al., J. Phys. Chem. B 2020, 124, 9009 - 9016

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